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Reanalyzing perfects

The definite article and the rise of Romance HAVE-auxiliaries

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Abstract

We propose a formal syntactic analysis of the Latin construction at the origin of Romance perfect tenses with a HAVE-auxiliary, arguing that the Latin construction ‘*habere* + NP.ACC + PaPa.ACC’ is a grammatically passive structure in which *habere* ‘have’ is the main verb, selecting a small clause with the accusative NP as subject and the participle as predicate. Drawing on Myler’s (2016) analysis of HAVE and insights from the literature on stative passives and the internal structure of participles, we show that the participles in these constructions may denote event and state tokens, and that different semantic readings of the construction correspond to a single syntactic structure. On the basis of this analysis, we develop a new account of the rise of analytic perfect tenses in Romance. We argue that the trigger for reanalysis was an independent change in the nominal domain — the emergence of a D-layer and in particular the grammaticalization of definite articles — which forced reanalysis specifically in contexts involving inalienably possessed direct objects. Once established in this critical context, the construction spread to new environments, including, for the first time, intransitive verbs.

Keywords: perfect, HAVE-stative, definite article, grammaticalization, possession, Latin, Romance

1 Introduction

Contemporary Romance languages have a periphrastic present perfect with an auxiliary descended from Latin *habere* ‘have’ and a past participle (henceforth: PaPa),¹ illustrated in (1), with possibly a secondary BE-auxiliary for unaccusatives. In what follows, we will refer to these constructions as HAVE-perfects.²

- (1) a. a chanté [French]
has sung
b. ha cantado [Spanish]
has sung

There is general agreement in the literature that the origin of this construction is found in Latin structures like (2), which feature a form of *habere* ‘have’, an NP marked for accusative case (NP.ACC), and a PaPa which agrees in number and gender with the accusative NP (PaPa.ACC).³ In the remainder of this article, for reasons that will be clarified in due time, we will use the term HAVE-stative to refer to this construction.

- (2) qui lippitudine oppressos
REL.NOM.M.SG inflammation.ABL(.F).SG overwhelmed.ACC.M.PL
habet oculos carnis
have.PRS.3SG eye.ACC(.M).PL flesh.GEN(.F).SG
‘whose fleshly eyes are suffering from an inflammation’ (Caesarius of Arles,
Sermo 104.1, ca. 520 CE)

The development of Latin *habere* into Romance HAVE-auxiliaries is one of the most often discussed case studies of grammaticalization⁴, featuring already in Meillet’s seminal paper on the topic (Meillet 1912/1982, 142–143). In this paper, we will not primarily dwell on the semantic side of the issue, but focus more narrowly on the syntactic corollaries of the transition from main verb to auxiliary.⁵ The key contribution of this paper is the identification of the conditions and the grammatical context in

¹Exceptions to this generalization are Portuguese and Galician, which use auxiliaries derived from Latin *tenere* ‘hold’ for the roughly equivalent construction. The use of TENERE-perfects is however more restricted (see, e.g., Harre 1991; Amaral and Howe 2012).

A note on abbreviations: we adhere to the Leipzig Glossing Rules; specifically, this paper uses the following abbreviations in glosses: 1 first person, 2 second person, 3 third person, ABL ablative, ACC accusative, DAT dative, DEM demonstrative, DET determiner, F feminine, FUT future, FUTPRF future perfect, GEN genitive, IMP imperative, INDEF indefinite, INF infinitive, IPFV imperfective, LOC locative, M masculine, N neuter, NOM nominative, PASS passive, PL plural, PLPRF pluperfect, POSS possessive, PRF perfect, PRO pronoun, PRS present, PRT particle, PTCP participle, Q interrogative, REFL reflexive, REL relative, SBJV subjunctive, SG singular, SUP superlative.

²These forms do not all have the same values in Romance, see, e.g., Schaden (2021) for a recent overview, and see also footnote 6.

³The literature on the prehistory of Romance HAVE-perfects is particularly rich, and cannot possibly be done full justice to in the present context. Among recent studies we may cite Rosén (1980), Salvi (1982), (1987), Vincent (1982), Pinkster (1987), Schwegler (1990, 117–123), Fleischman (1993), Jacob (1995a), (1995bb), (1998), Napoli (2007), Cennamo (2008), Târa (2012), (2014), Wehr (2012), Hertenberg (2015), Haverling (2016), Pieroni (2016) and Glasenapp (2024).

⁴We define the term ‘grammaticalization’ as the genesis of a novel piece of functional morphology, out of a pre-existing lexical item (primary grammaticalization) or out of another functional item (secondary grammaticalization).

⁵We have dealt with the semantics of Latin HAVE-statives in detail in Danckaert and Schaden (To appear).

which the *primary*⁶ grammaticalization (see, e.g. Narrog and Heine 2021) of *habere* + PaPa into a present perfect construction came to pass.

1.1 HAVE-perfects from Latin to Romance: Syntactic discontinuities

Syntactic (or structural) reanalysis has often been invoked to model the rise of the Romance HAVE-perfects (see, among many others, Ramat 1983, 1984; Pinkster 1987).⁷ In the most general case, this process can be said to obtain under the following conditions. Assume a communicative situation with a speaker (sender) S and a hearer (receiver) H. When S produces an utterance U, they construe a mental representation of U, which contains, among other things, information about U’s hierarchical structure (that is, a syntactic parse). Upon receiving U, H construes a syntactic structure of their own by parsing the incoming linear string. Whenever there is a discrepancy between the parses assumed by S and H, we say that there is reanalysis. Note that the possibility for reanalysis to occur is conditional upon the potential for U to be structurally ambiguous, that is, compatible with more than one structural analysis. In the particular case when reanalysis yields a structure which cannot be generated by the mental grammar of S, H has effectively innovated a new structure. More precisely, syntactic change has occurred at the level of H’s I-language. Whenever H subsequently produces an utterance which unambiguously features this new structural possibility, the change may further spread across the speech community, yielding syntactic change at the level of E-language. In the generative literature on syntactic change, it is standardly assumed that syntactic change through reanalysis happens intergenerationally, that is, in a communicative setting in which H is an L1-learner.

In our specific case, according to Pinkster (1987, 196), the Latin input and Romance output structure (that is, before and after reanalysis) look like (3) (= Pinkster’s (4)):

$$(3) \quad habere + (\text{Object} + \text{participle}) \rightarrow (habere + \text{participle}) + \text{Object}$$

(3) indicates that in Latin, the PaPa is more closely connected to the NP.ACC, whereas in Romance, it is grouped together with *habere*. There are of course various ways in which the notions of being ‘connected’ and ‘grouped together’ can be further formalized (e.g. in terms of phrase structure or dependencies), but in essence, most extant accounts assume some version of this general development.

⁶We are aware of the rich literature on the *secondary* grammaticalization of perfects (see, e.g., Harris 1982; Bybee and Dahl 1989; Schaden 2012; De Mulder and Patard 2020), but we will focus here only on the point at which *habere* transforms for the first time into a TAM-auxiliary.

⁷A rather different line of analysis capitalizes on the role of (stative-inactive) argument alignment in Late Latin: see La Fauci (1997, 2005, 2006), Cennamo (2008) and Ledgeway (2012, 317–326; 341–349). In short, this line of reasoning is based on the assumption that from Archaic Latin onwards, BE-auxiliaries in analytic perfects only ever occur with passives and deponents, that is, when the syntactic subject is underlyingly an internal argument; the rise of HAVE-auxiliaries is argued to be the reflex of an analogical extension of this pre-existing stative-inactive alignment system (Ledgeway 2012, 317–318). Despite there being some evidence that the grammar of Late Latin shows traces of stative-inactive alignment (Korkiakangas 2016), we do not consider it likely that this was causally related to the rise of HAVE-perfects. In particular, the assumption that “the overwhelming majority of deponents are unaccusative predicates” (Ledgeway 2012, 316) strikes us as entirely unsubstantiated.

Unfortunately, we are not aware of any comprehensive analysis of how and why reanalysis took place. A very general problem is that reanalysis may actually be too powerful a mechanism, if not constrained in principled ways: if all that is needed for rebracketing to take place is the presence of structural ambiguity (which is rampant in natural languages), why does syntactic change not happen more frequently? Therefore, in order for any diachronic account based on reanalysis to be convincing, one needs to spell out in full detail the conditions under which this process is *predicted* to occur.⁸ In other words: what are the necessary and sufficient conditions for rebracketing to take place, and in which grammatical context do these conditions obtain? Answering these questions is the major aim of this paper: in section 5.2, we will propose a specific scenario to explain why reanalysis took place, with the genesis of HAVE-perfects in Romance as a result.

If the transition from HAVE-statives to HAVE-perfects crucially involved syntactic reanalysis (as argued, e.g., by de Acosta 2011), the change must have been abrupt. This goes against what is perhaps the ‘standard view’ of the grammaticalization of the Romance HAVE-auxiliary, which emphasizes the gradual reduction of the possessive meaning associated with HAVE: see, among many others, Squartini and Bertinetto (2000, 405), Drinka (2007, 109) and De Mulder and Patard (2020, 1497). We take it that the semantics of *habere* was highly abstract right from the outset, devoid of much lexical content that could be bleached away (for further details, see Danckaert and Schaden To appear).⁹ This is not to say that we do not believe that certain aspects of the entire development may have been gradual. The genesis of HAVE-perfects entailed a reorganization of the overall tense-aspect system of Late Latin/Early Romance, to the effect that the new periphrastic structures took over some functions of the pre-existing synthetic perfects. This process of competition obviously did not unfold overnight (it may in fact last until today). Nevertheless, the fact that the spread and secondary grammaticalization of HAVE-perfects may have been gradual does not preclude that their genesis was instantaneous.

Assuming that reanalysis is indeed a crucial ingredient of the analysis, it remains to be determined what the exact properties are of the type of input string which L1-learners parsed differently than their care-givers. While there is no off-the-shelf solution to this problem, in the existing literature on the history of Romance HAVE-perfects, there is a wide consensus that in order for a given surface string uttered

⁸More fundamental concerns have at times been raised regarding the necessity and explanatory power of reanalysis as a crucial component of syntactic change. For example, Haspelmath (1998) argues that this mechanism is not needed to explain processes of grammaticalization, and that it is only involved in a relatively small number of other instances of syntactic change (see, among others, Whitman 2000 for comparable considerations). De Smet (2009) goes even further, by claiming that diachronic accounts based on reanalysis are logically inconsistent, and that the mechanism is to be dispensed with altogether (at least as a primitive notion). Here we adopt the position that if sufficiently motivated, calling upon reanalysis to analyse syntactic change may indeed meet the required standards for explanatory adequacy. One important attempt at providing a theoretical basis for the phenomenon of reanalysis can be found in Lightfoot (1979), who invokes the ‘Transparency Principle’ to predict under which conditions reanalysis should occur. Nevertheless, by the author’s own admission, this principle is not explicitly formalized (Lightfoot 1979, 344), and it is not immediately clear to us how we could apply it to the case study at hand.

⁹The idea that the semantics of non-auxiliary HAVE is very abstract was also proposed in Ramchand (2014, 224–227). The author draws evidence in favour of this claim by considering (synchronically stable) patterns of polysemy across a range of complementation frames in which English *have* acts either as a full verb or as a light verb. On ‘light verb’ usages of Latin *habere*, see Danckaert and Schaden To appear, Section 2.1. We thank an anonymous reviewer for drawing our attention to Ramchand’s work.

in a particular context to qualify as a potential source construction of a periphrastic perfect, the subject of *habere* needs to be coreferential with the understood agent or experiencer participant associated with the past participle. This condition is fulfilled in (4), but not, e.g., in (2).

- (4) Ioseph [...] *suspectum* *habebat* *uitium*
 Ioseph.NOM(.M).SG suspected.ACC.N.SG have.IPFV.3SG flaw.ACC(.N).SG
 ‘Joseph had a suspicion of sin.’ (Ambrose, *Epistulae* 8.56.13, 380 CE)

We will call this first necessary condition for reanalysis the *coreference condition*. This requirement, however, conceals the presupposition – often left implicit in the literature – that the PaPa needs to have at least enough verbal functional structure to accommodate an agent argument.¹⁰ This is the second necessary condition, which we will call the *verbal PaPa condition*.¹¹

While these two necessary conditions seem to be generally agreed upon, the status and analysis of structures like (4) are disputed. Whatever the correct analysis, such structures have been available from Archaic Latin onwards, as witnessed by the example from Plautus in (5) (for further discussion of this pattern in Plautus, see Happ 1967, 92–104).

- (5) ut *eam* *iunctam* bene *habent* inter se
 how DEM.ACC.F.SG joined.ACC.F.SG well have.PRS.3PL between REFL.ACC
 ‘how well they foster it (viz. their mutual friendship) among themselves’
 (Plautus, *Cistellaria* 25, ca. 208 BCE)

¹⁰Note that what formally looks like a PaPa may in fact be a plain adjective devoid of verbal functional structure. Consider, for example, the two examples in (i):

- (i) a. eos libros, qui ab illo editi
 DEM.ACC.M.PL book.ACC(.M).PL REL.NOM.M.PL by DEM.ABL.M.SG published.NOM.M.PL
 sunt
 be.PRS.3PL
 ‘those books which were published by him’ (Hegemonius, *Acta Archelai* 68.5, ca. 350 CE)
 b. editi colles
 high.NOM.M.PL hills.NOM(.M).PL
 ‘high hills’ (pseudo-Charisius, *Synonyma Ciceronis* p. 433, l. 30 (ed. Barwick), ca. 8th c. CE)

The form *editi* ‘published’ in (ia) can plausibly be interpreted as a truly verbal PaPa derived from the base verb *edere* ‘emit, produce, raise’, forming a periphrastic passive in combination with the BE-auxiliary *sunt*. For one thing, the clause contains a *by*-phrase, revealing the presence of an agent, and by this token, the fact that the PaPa has the argument structure of a verb. In contrast, the homophonous element *editi* in (ib) is in all likelihood a pure adjective: though it is formally identical to the verbal PaPa in (ia), its lexical meaning is no longer fully transparently related to that of the verb *edere*; the most natural interpretation of (ib) is not one according to which the elevated position of the relevant hills is presented as the result of an act of lifting up (volitional or otherwise).

The contrast between the two items (i) may seem fairly clear, but in many other cases, it is much less obvious to tell apart verbal PaPas from homophonous adjectives. We will further pursue this discussion in section 2.2.

¹¹It has occasionally been suggested that there is yet another condition which the Latin source construction of Romance HAVE-perfects needs to fulfil, which concerns the lexical aspect of the PaPa. Concretely, it has been argued that this should be derived from a telic verb, as this type of predicate naturally implies a resultant state (see, among others Napoli 2007, 8; compare also Lucot 1940 and Rosén 1980). However, we believe that although this line of reasoning is essentially on the right track, it is not necessary to postulate a telicity condition distinct from that requiring the presence of an agentive event. Instead, as we will argue in section 2.3, we take it that the participial morphology introduces a post-state, which in combination with the agent participant introduced lower in the structure effectively derives the same effect.

According to Hertzenberg (2015), from Archaic Latin onwards, at least some structures of this type qualify as true HAVE-perfects¹² – whose interpretation is, for all intents and purposes, similar to that of HAVE-perfects in, e.g., contemporary French and Spanish. In contrast, authors like Thielmann (1885) and Pinkster (1987) deny that early or Classical Latin had full-fledged present perfects, but claim that the first real perfects appear in Late Latin. The most conservative position is held by scholars like de Acosta (2011), Fruyt (2011), Adams (2013) and Țăra (2014), who claim that HAVE-perfects are altogether absent throughout Antiquity, and that periphrastic perfects with HAVE only developed at the very earliest during the transition from Latin to Romance (see also Herzog 1910).

The strongest argument in favour of the latter position – which we will endorse in this paper – is the observation that no HAVE-perfects of intransitive verbs are attested before 900 CE (cf. Herzog 1910, 177–185). In contrast, Latin HAVE-statives invariably feature a PaPa derived from a transitive base verb. If true HAVE-perfects existed in Latin, the restriction to transitive predicates would be entirely mysterious. To the best of our knowledge, Latin HAVE-statives did not undergo any significant change during the extended period from around 200 BCE until the eighth century CE (that is, until right before the individual Romance daughter languages of Latin manifest themselves as independent vernaculars), which raises the question how the grammaticalization of HAVE-auxiliaries was triggered. This is the main reason why we locate this development after 750 CE, that is, in a period from which few written records are preserved which may reflect the spoken language (written Romance data are not yet available, and written Latin reflects the learned language of the so-called ‘Carolingian renaissance’, which started towards the end of the eighth century).¹³ Nevertheless, the observation that HAVE-perfects were available in all the early Romance daughter languages of Latin suggests that the phenomenon was a shared inheritance, whose origins are to be traced back to a stage in which the E-language Latin was still sufficiently unified in order for each of the individual daughter languages to inherit HAVE-perfects. Although our analysis does not exclude that HAVE-auxiliaries were innovated in parallel in independent pockets of Early Romance, this scenario strikes us as less plausible.

1.2 The idea in a nutshell: Definiteness, inalienable possession, and HAVE-constructions

The two conditions (coreferentiality and verbal PaPa) we have encountered so far are only necessary conditions for HAVE-perfects to be innovated. What is lacking is one or more sufficient conditions which predict that (some) HAVE-statives should change into HAVE-perfects; as far as we are aware, no convincing such condition has been identified in the literature, and indeed, most existing analyses stop short of trying to model the precise conditions under which HAVE-perfects were first innovated. The

¹²We take ‘true (HAVE-)perfects’ in the remainder of the article to be constructions with a HAVE-verb where the coreference and the verbal PaPa conditions hold necessarily, imposed by the construction’s grammar.

¹³Hertzenberg (2015) similarly locates the actuation of the change in a period without good textual evidence, namely before the appearance of the earliest substantial records of Latin. Under this approach, it remains to be explained why competition between synthetic perfects and transitive HAVE-perfects should have remained stable for a period of *ca.* 1000 years, to be resolved very quickly afterwards.

main purpose of our paper is to develop an explanatory analysis which predicts that in a particular set of syntactic contexts, HAVE-perfects should necessarily emerge. In order to do so, we identify one additional necessary condition and – crucially – one sufficient condition that needs to be fulfilled in order for a given HAVE-stative to qualify as a potential source of a true perfect. The upshot is that only a very specific subset of HAVE-statives which satisfy the two necessary conditions discussed above qualifies as the source construction of the Romance periphrastic perfect.

The intuition behind our proposal is based on the observation that Latin HAVE-statives live on in two guises in Romance (as pointed out by De Mulder and Patard (2020), who themselves refer to Detges 2000, 2006): one is the periphrastic perfect (6a), the other is a structure much more akin to the Latin source construction (6b) (examples taken from De Mulder and Patard 2020, 1496). The pair in (6) features examples from Old French, but the two construction types live on until this day (in Modern French, cf. (7), as well as in other Romance varieties). Ideally, an explanatory account of the genesis of Romance HAVE-perfects should account for how two distinct structures emerged out of the same Latin source.

- (6) a. Li quens Rollant, il l’=ad e prise e fraite.
the count Roland he it=have.PRS.3SG and taken.F and destroyed.F
‘Count Roland conquered it and destroyed it.’ (*Roland*, v. 1555, ca. 1100)
- b. Escababi i ad le *chef trenchet*.
Escababi he have.PRS.3SG the head cut
‘Escababi had his head cut off.’ (*Roland*, v. 663, ca. 1100)
- (7) a. Les élèves ont levé [la main].
the.PL pupils have.PRS.3PL raised the.SG hand.SG
‘The pupils have raised their hand.’
- b. Les élèves ont [la main] *levée*.
the.PL pupils have.PRS.3PL the.SG hand.SG raised.F
‘The pupils have their hand in the air.’

One important observation is that the Romance b-examples express inalienable possession (roughly, a possessive relation in which the possessed entity is construed as inherently or permanently associated with the possessor; see section 3). This pattern is one member of the wider family of stative constructions which appear to be fairly close in meaning to the pre-existing Latin HAVE-statives. Other cases exemplifying this pattern, such as (8), do not involve inalienable possession:¹⁴

- (8) Nous avons la *voiture* [*garée* devant la maison].
we have the car parked in front of the house
‘Our car is parked in front of the house.’

Despite the existence of structures like (8), we will argue that only HAVE-statives expressing inalienable possession like (7b) and (6b) are directly related to the rise

¹⁴In Danckaert and Schaden (To appear, § 3.3.3–3.3.4), building on Iatridou (1996) we have discussed certain semantic restrictions on the subject and the predicate of small clauses under HAVE-verbs. For further discussion of HAVE-statives in the Romance language family, see, among others, Harre (1991) and Egerland (1996).

of auxiliary HAVE. Crucially, in Romance, inalienable possession typically features a semantically ‘light’ definite determiner.¹⁵ Capitalizing on this observation, for reasons that will be clarified in due course, we propose an additional necessary condition on Latin HAVE-stative as the source of Romance HAVE-perfects, according to which the NP.ACC must be inalienably possessed, as in (9). We will refer to this as the *inalienability condition*.

- (9) quando Moses *erectas* habebat *manus*
when Moses.NOM(.M).SG raised.ACC.F.PL have.IPFV.3SG hand.ACC(.F).PL
‘when Moses had his hands raised’ (Maximus Taurinensis, *Sermo* 45, l. 40
(Mutzenbecher), *ca.* 400 CE)

The combination of the coreference, verbal PaPa and inalienability conditions still is not sufficient, however, for syntactic change to emerge. We propose that in order for this to happen, argument nominals – and crucially those that are construed as inalienably possessed – need to be expanded with a D-layer. We will show that this happens towards the very end of the Late Latin period (at least as early as the 8th c. CE, and perhaps a bit earlier, see Varvaro 2013, 30ff.). While at that time, the use of definite articles had not yet spread to the entire range of contexts in which they are used in modern Romance languages, the important point for us is that for the expression of inalienable possession, the use of the definite article is found from the earliest (8th century) attestations on (see section 5.1), whereas unambiguous HAVE-perfects, in particular those featuring an intransitive PaPa, only appear in the individual Romance daughter languages of Latin.

As we will argue, the grammaticalization of definite articles entailed a new way of syntactically encoding inalienable possession relations. A side effect of this development is that a subset of the pre-existing surface strings could no longer be analysed as HAVE-statives, forcing learners to impose a parse on the relevant input strings that was not available to previous generations. By doing so, they innovated periphrastic perfects with auxiliary HAVE. In combination with the three necessary conditions identified above, the grammaticalization of definite articles is thus the sufficient condition leading to the grammaticalization of HAVE-perfects.

Our main claim is thus that two of the most salient grammatical developments that took place in the transition from Latin to Romance, namely the rise of definite articles and the emergence of HAVE-perfects, are not two independent innovations. Rather, we contend that it was the grammaticalization of the definite article that triggered the grammaticalization of HAVE-perfects, in the specific context of inalienable possession. In other words, the observed time lag between the first *bona fide* definite determiners and the first attested intransitive HAVE-perfects is no accident: rather, as a result of the emergence of a D-layer in the nominal domain, L1-learners were forced to reanalyse a subset of the pre-existing HAVE-constructions with a PaPa as ‘true perfects’. At a more general level, in contrast with studies that have focused on the lexical semantics of the PaPa (in particular Târa 2014), we contend that what is more crucial to the diachrony of the Romance periphrastic perfect is the interpretation of NP.ACC, that

¹⁵How a ‘semantically light’ determiner can be characterized in more precise terms will be examined in section 3.

is, the internal argument of the PaPa. We are not aware of any previous study which has pursued this line of analysis.

1.3 Roadmap

The remainder of this paper is structured as follows. In section 2, we offer a detailed description of the Latin source construction of the Romance periphrastic perfect. Based on, among others, Alexiadou et al. (2015); Maienborn et al. (2016), we also provide a syntactic analysis of the relevant construction. We then turn to the genesis of Romance perfects: why did reanalysis happen, to the effect that *habere* became an auxiliary? Section 3 lays out our background assumptions regarding the syntax and semantics of inalienable possession constructions, focusing in particular on the role of definite determiners. In section 4, we consider how inalienable possession was expressed in Latin, a language which did not have definite articles. Section 5.1 is concerned with the relative chronology of the rise of definite determiners and HAVE-perfects in Romance: the extant evidence clearly shows that the former pre-dates the latter. Finally, in section 5.2 we argue that once definite articles had come into being, it is predicted that as an indirect result, the periphrastic perfect should be innovated too. Section 6 sums up the main findings of the paper, and it briefly explores some potential extensions of our analysis that go beyond the Romance language family.¹⁶

The upcoming analysis of Latin HAVE-statives is based on examples drawn from the online repository of Latin texts available at www.brepolis.net. As mentioned, we assume that HAVE-perfects were innovated late: for this reason, we will mainly focus on data from Late Latin, which we loosely define as the period from around 200 CE until the early eighth century CE.¹⁷ Needless to say, during this extensive period of over 500 years the Latin language underwent a wide variety of changes, but crucially, to the very best of our knowledge, the syntax of HAVE-statives, as well as the properties of the passive past participles that can be used in this construction, remain by and large constant.

Given the timing of the grammaticalization of the TAM-auxiliary HAVE which we assume, there is no (direct) textual evidence bearing on this phenomenon. As a result, the only way in which progress can be made is by making reasonable inferences about how and when the process may have unfolded. By necessity, this way of reasoning involves some amount of syntactic reconstruction. Methodologically, our endeavour

¹⁶A development that we will not be concerned with is the grammaticalization of *habere* into future and conditional affixes. There are many reasons to believe that this process is entirely unconnected to the grammaticalization of HAVE-auxiliaries in perfects. First, the timing of the two developments is entirely different: on the one hand, *habere* with a past participle is well attested from Archaic Latin (basically the writings of Plautus, Terence and Cato) onwards, whereas up until the (early) third c. CE, *habere* with an infinitive remains extremely rare (the pattern is however well attested in many Late Latin texts); on the other hand, the earliest water-tight attestation of the grammaticalized synthetic future dates from the second half of the sixth c. CE (Ledgeway 2012, 136–137), whereas there is no Latin evidence for a grammaticalized HAVE-auxiliary: as mentioned, the first unambiguous attestations of this pattern are found in the individual Romance daughter languages. Second, in contexts where *habere* is complemented by an infinitive, it grammaticalizes to become a bound affix, whereas in the perfect, the outcome of grammaticalization is a free-standing auxiliary, whose phonological shape remains identical to that of non-auxiliary usages of the same HAVE-verb (at least in those Romance varieties in which the latter are still available).

¹⁷This corresponds to the *Aetas patrum* from the Brepolis-database, which includes texts from 200 until 735 CE.

is relatively modest in scope, and involves two major components, namely (i) theoretical generalizations about syntactic structures; and (ii) generalizations about cross-linguistically recurrent (morpho-)syntactic changes, such as grammaticalization paths. We take it that these elements constitute a sufficiently solid basis to bridge the gap between Latin structures like (2) and Romance periphrastic perfects like (1).

2 The syntax of Latin HAVE-statives

In this section we offer a detailed syntactic analysis of the source construction of the Romance HAVE-perfects.¹⁸ In section 2.1, we start by pointing out that a number of syntactic parses are available for the surface string ‘*habere* + NP.ACC + PaPa.ACC’, and that only one of them is relevant for the diachronic development at hand. We argue that in the relevant structure, *habere* selects a small clause complement in which NP.ACC is the subject and PaPa.ACC the predicate. We also show that this analysis can successfully account for Latin examples that have been analysed as true perfects. We proceed in section 2.2 to analyse the internal structure of PaPas occurring in HAVE-statives, focusing in particular on how much verbal functional structures these elements may have. In section 2.3, we combine these elements to provide a full analysis of the entire construction.

2.1 A small clause structure for Latin HAVE-statives

In section 2.1.1, we recapitulate de Acosta’s argument that the surface string is ambiguous between several readings, not all of which are relevant for the genesis of Romance HAVE-perfects; in section 2.1.2, we argue that it is unwarranted to assume any grammatical difference between de Acosta’s *Attained State* and *Affectee*-readings; finally, in section 2.1.3, we make a case for assuming a small-clause structure for the complement of HABERE in Latin HAVE-statives. We have analysed the structure of Latin HAVE-statives in more detail in Danckaert and Schaden (To appear, § 3, especially 3.3): here we only report on the major conclusions reached there.

2.1.1 Disambiguating *have*-PaPa strings

According to de Acosta (2011), the string ‘*habere* + NP.ACC + PaPa.ACC’ can instantiate three different constructions. In the first, the PaPa functions as a modifier in a complex NP which as a whole acts as the direct object of the HAVE-predicate.¹⁹ De de Acosta refers to this pattern as the *Adnominal type*. This analysis imposes no *a priori* restrictions on the internal structure of the PaPa, which may either be a pure adjective, or alternatively, a reduced relative clause in which the participle has some verbal

¹⁸A milestone in the study of the Latin origins of Romance HAVE-perfects is Thielmann (1885), a monumental article which contains a plethora of examples of *habere* combined with a PaPa, dating from early Republican to (very) Late Latin. Further useful data are collated in Bulhart (1936) (see in particular pp. 2425.33–2427.39; 2429.69–2430.9; 2452.67–2453.75). Most recent analyses of the same topic – with the notable exception of Tăra (2014) – are based on the empirical documentation provided in these two studies.

¹⁹According to Jacob (1995a, 367), this configuration is the basic structure of all Latin precursors of Romance HAVE-perfects.

functional structure. This is illustrated in (10), where the PaPa *captum* ‘caught’ modifies the noun *piscem* ‘fish’ (for arguments why this example unambiguously features an adnominal PaPa, see Danckaert and Schaden To appear, § 3.3.4).

- (10) Merito habemus [primum *piscem*
consequently have.PRS.1PL first.ACC.M.SG fish.ACC(.M).SG
[hamo *captum*]].
fish-hook.ABL(.M).SG caught.ACC.M.SG
‘Consequently, we have (here) the first fish caught with a fish-hook.’ (Augustine, *Enarrationes in Psalmos* 137.16, ca. 400 CE)

In (10), there is a thematic relation between *habere* and the NP headed by *piscem*. Given this analysis, if NP.ACC is pronominalized, it should not be possible for the PaPa to be maintained. Conversely, we predict that whenever we encounter the string ‘*habere* + pronoun.ACC + PaPa.ACC’ (as in example (11) below), the complex NP analysis should not be tenable.

There is every reason to believe that in the history of Latin/Romance, adnominal PaPas (reduced relatives) have been available from Archaic Latin until the present-day Romance languages; to the best of our understanding, this phenomenon has not undergone any major structural change that could have led L1-learners to reanalyse a complex NP with a participial modifier in the complement position of a *have*-verb as a periphrastic perfect. Therefore, we assume that this construction is orthogonal to the diachrony of Romance (HAVE-)perfects. In what follows, we will no longer be concerned with it.

In the two other structures distinguished by de Acosta (2011), NP.ACC and the PaPa.ACC form a small clause constituent rather than a complex NP.²⁰ At an interpretive level, two subtypes can be distinguished. The first is what de Acosta refers to as the *Attained State reading*. Here, the subject of *habere* is the understood agent of the verbal process denoted by the PaPa. In (11), the person who accepted to act as a lawyer in a particular court case (cf. *susceptam*) is most naturally interpreted as being identical to the one who at the reference time was one of the official parties in the relevant case.

- (11) in quadam actione, *quam* habuit cum
in INDEF.ABL.F.SG case.ABL(.F).SG REL.ACC.F.SG have.PRF.3SG with
Cicerone *susceptam*
Cicero.ABL(.M).SG undertaken.ACC.F.SG
‘in some court case, which he (viz. Hortensius) had undertaken with Cicero’
(Macrobius, *Saturnalia* 3.13.3, ca. 431 CE)

The θ -role of the subject of *habere* in this reading is what de Acosta (2011, 165) calls an *attainer*: the subject is the holder of a state (as is generally the case with *habere*) and thus in some ways experiencer-like; but importantly, this state has been brought

²⁰We use the term ‘small clause’ informally, to refer to a non-finite predicative nucleus featuring a (pro)nominal subject and an AP, NP or PP as the predicate (without any overt copular element). As we will explain below (see in particular section 2.4), we assume the syntactic realization of this phenomenon to involve a Predication Phrase (PredP, in the tradition of Bowers 1993). On the small clause analysis of complements of Latin *habere*, see also Salvi (1987).

about through the subject’s own agency. In other words, the subject of *habere* is also *pragmatically*²¹ identified with the initiator of the event denoted by the PaPa (in (11), the taking up of a court case). The label ‘attainer’ captures this dual role: the subject has actively brought about the state they now possess or experience. In examples like these, the NP.ACC can be replaced by a (demonstrative, interrogative or relative) pronoun (e.g. *quam* ‘this’ in (11)), while maintaining the PaPa.ACC: this precludes an analysis whereby the PaPa acts as a reduced relative.

The final reading distinguished by de Acosta is what he calls the *Affectee interpretation*. Contrary to the Attained State reading, in this case, the subject of *habere* is not identified with the agent of the PaPa. For example, in (12) there is an explicit *by*-phrase specifying the agent, namely *a Deo Patre* ‘by God the Father’, which does not corefer with the subject of *habere*.

- (12) *habebit* *eam* *datam* *a Deo*
have.FUT.3SG DEM.ACC.F.SG given.ACC.F.SG by god.ABL(.M).SG
Patre
father.ABL(.M).SG
‘He will have it (viz. the grace of the Holy Spirit) as a gift from God the Father’
(Jerome (translator of Didymus Alexandrinus), *Liber de Spiritu Sancto* 16,
ca. 387 CE)

In this reading, the subject of *habere* undergoes a situation not of their own making, and usually, this happens involuntarily. The θ -role of the subject of *habere* is a non-agentive experiencer (which may be a beneficiary, a sufferer, recipient, etc.), but never an agent, cause, patient, or theme.

2.1.2 Are Attained State and Affectee interpretations *grammatically* different?

Although we concur with de Acosta (2011) that the Affectee and the Attained State reading are not to be conflated (only the latter is relevant for the grammaticalization of HAVE-perfects), we see no compelling reason to assume that the two types have to differ in terms of their grammatical structure (that is, syntactic structure and/or [non-lexical] semantics). Rather, we take it that the distinction between the two readings is pragmatic in nature (according to de Acosta (2011), the θ -roles of the participants involved in the Attained State and Affectee readings are in complementary distribution).

This conclusion is supported by the observation that there are no principled restrictions on the nature of the non-finite predicate that can appear in this environment: the relevant item can be a plain adjective, a PaPa with some verbal structure, a PP or even an NP. Consequently, the source construction of Romance HAVE-perfects is merely a special case of a broader phenomenon in which *habere* takes a small clause complement. To appreciate this point, consider the examples in (13), featuring the

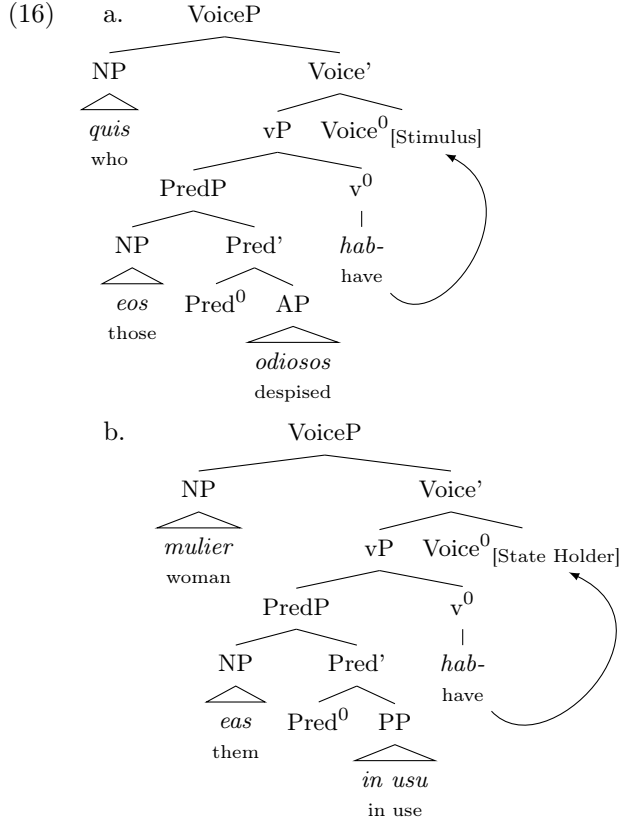
²¹Note that de Acosta (2011) would not necessarily commit to the idea that this is a pragmatic inference, but we have argued in Danckaert and Schaden (To appear) that it is.

adjective *odiosos* ‘offensive’ instead of a PaPa, (14) with the NP *diuersorium* ‘inn’ in the same slot, and (15) with a PP.²²

- (13) Aut quis non habeat *odiosos*
or who.NOM.M.SG not have.PRS.SBJV.3SG offensive.ACC.M.PL
eos, qui respuunt quidem constituta
DEM.ACC.M.PL REL.NOM.M.PL disparage.PRS.3PL PRT laws.ACC
patrum [...]?
fathers.GEN
‘Or who wouldn’t despise those who disparage the laws of our fathers?’
(*Historia ecclesiastica tripartita* 5.30.1, ca. 550 CE)
- (14) ager, quem [NP *diuersorium*] habuerit
field.NOM(.M).SG REL.ACC.M.SG inn.ACC(.N).SG have.FUTPRF.3SG
praedictus
aforementioned.NOM.M.SG
‘the farm, which the aforesaid <governor> used as a lodging place’ (*Codex Theodosianus* 23.3.69.8, 369 CE, translation adapted from Pharr (1952, 29))
- (15) quamuis *eas* mulier [PP *in usu*]
although DEM.ACC.F.PL woman.NOM(.F).SG in use.ABL(.M).SG
habeat
have.PRS.SBJV.3SG
‘despite the fact that the wife has them in use’ (*Digesta seu Pandectae Iustiniani Augusti* 23.3.69.8, ca. 530 CE)

In (16a-b), we give simplified analyses of examples (13) and (15) (we abstract away from linear word order; the *wh*-subject in (16a) is represented *in situ*). The basic structure is the same in both cases, featuring a verb *hab-* combined with a PredP, but the complements of the respective Pred-heads are not the same (AP in (16a) vs. PP in (16b)). Notice also that in these constructions without PaPas, the semantic content and the θ -role assigned to the subject in SpecVoiceP are not identical. We take this to indicate that the opposition between Attained State and Affectee readings is not due to different syntactic configurations in HAVE-statives, but that there is a more general issue of θ -role diversity in this construction.

²²Bulhart (1936) cites additional examples of *habere* complemented by a small clause with a nominal (pp. 2423.27–2424.41), adjectival (pp. 2424.42–2425.32) and PP (p. 2427.40–2428.37) predicate.



Importantly, in these structures NP.ACC (*eos* in (16a) and *eas* in (16b)) is in no way thematically related to the HAVE-verb, but only case marked by it (presumably in a type of Exceptional Case Marking (ECM) configuration).²³ In the particular case of HAVE-statives, the subcategorization frame of *habere* is such that it requires a state holder as the external argument (the latter is typically – but not always – a sentient being), and a non-finite clausal complement (the small clause denoting a [resultant] state) as its internal argument. NP.ACC, on the other hand, acts as the subject of the small clause predicate. As we will see in section 2.2, if the latter is a passive PaPa, NP.ACC can be qualified as a derived subject, that is, an internal argument which typically carries the thematic role of Theme or Patient.

²³Egerland (1996, 121–123) puts forward three arguments against an ECM analysis of HAVE-statives, none of which strike us as convincing. Let us briefly comment on two of his objections. First, the idea that HAVE-verbs can never assign accusative is difficult to reconcile with the observation that in Latin, *habere* consistently appears with an accusative-marked direct object. If *habere* is ruled out as a source for this case marking pattern, it is unclear what the alternative case assigner would be. Second, the fact that HAVE-statives resist passivization should not be taken to indicate that the active baseline pattern does not have ECM syntax, given that there is ample evidence for a double dissociation between ECM and passive raising. Moreover, possessive HAVE tends to resist passivization in many languages, and Latin does not appear to be an exception in this regard (passive *haberi* usually means ‘be considered’).

2.1.3 Why a small-clause complement to *habere*?

Several analyses have been proposed for (subsets of) the Latin source construction of Romance HAVE-perfects. Up to now, we have simply presupposed that the small-clause analysis is correct, without providing any evidence. The conclusion from Danckaert and Schaden (To appear) is that a small-clause analysis can cover all relevant cases, and is the only analysis to do so.

Crucial evidence comes from cases where *habere* appears with a complement *clause* and a predicate that is not a PaPa, such as (17).

- (17) [Quid [...] temere factum dicatur]
 Q.NOM.N.SG recklessly done.NOM.N.SG say.PRS.SBJV.PASS.3SG
 incertum habeo.
 uncertain.ACC.N.SG have.PRS.1SG
 ‘It is unclear to me what is said to have been done recklessly.’ (Ambrose, *Epistulae* 10.76, 386 CE)

For (17), only a small clause parse seems available. First, adjectives such as *incertum* ‘uncertain’ in (17) never act as attributive modifiers of clauses. Second, an analysis with a three-place version of *habere* along the lines of Hertenberg (2015) is excluded, as this would require *habere* to take an embedded question as its internal argument, which is generally impossible (see Pinkster 1987). Third, given the absence of a PaPa, parsing *habere* as an auxiliary is also not possible.

Furthermore, we have argued in Danckaert and Schaden (To appear, § 3) that combinations of ‘*habere* + PaPa + NP.ACC’ like (18) that were analysed as true HAVE-perfects in Hertenberg (2015) are also amenable to a small clause analysis.²⁴

- (18) *quid* Athenis *exquisitum* habeam
 what.ACC.N.SG Athens.LOC(.F).PL found.out.ACC.N.SG have.PRS.SBJV.1SG
 ‘what I have found out in Athens’ (Cato, *ad Filium* fr. 1, ca. 175–150 BCE)

Summing up, our argument is essentially one of explanatory parsimony. Given examples like (17), we need to assume that HABERE could take a small clause as an argument. Furthermore, since we can explain Attained State and Affectee interpretations as instances of *habere* with a small clause complement, we should do so, unless the data forces us to analyse these structures differently, e.g., as true HAVE-perfects.

Having analysed the gross constituency of Latin HAVE-statives, we now turn to the internal structure of the PaPa.

2.2 The internal structure of the PaPa

In this section we focus on the properties of PaPas occurring in Latin HAVE-statives, building on the recent literature on the structure of participles (cf., e.g., Anagnostopoulou 2003; Alexiadou et al. 2015; Gehrke 2015; Berro 2019; Salzmann and Schaden 2019; Bešlin 2023). This body of literature on PaPas (which is centered on passive

²⁴ Anticipating the analyses in Sections 2.2–2.3, the idea is that *quid* ‘what’ is base-generated as the internal argument of the verb within the projection of the participle, after which it moves to the specifier of PredP (to undergo *wh*-movement at the final stage of the derivation).

constructions) is mainly preoccupied with the following questions: i) how much (functional) structure is included under the PaPa; ii) does the PaPa denote an event-type, or an event-token (intuitively: is a PaPa like *danced* about a single, existing event of *dancing*, or about something more abstract)? iii) what is the status of event modifiers (incorporated or not)?

Each of these questions essentially addresses the same underlying issue: to what extent does the participle qualify as verbal or adjectival? As mentioned, this issue is highly relevant, since scholars committed to the idea that there were real perfects in Latin are *ipso facto* committed to the idea that the PaPas in these formations are completely verbal (and thus, contain a great amount of functional structure, refer unproblematically to event-tokens, and do not have incorporated event modifiers).

We will address each of these questions in turn. For reasons mentioned in the previous section, we are working under the hypothesis that Attained State and Affectee-type readings are syntactically identical. However, there is no guarantee that what formally looks like a morphologically homogeneous class of PaPas may not in fact correspond to a family of structurally distinct formations. One class of participles that is likely to be different is that of deponent PaPas, which can have full-fledged perfect meanings, and which we will not deal with here.²⁵ However, even among *bona fide* passive PaPas, some forms may be endowed with more functional structure than others (Kratzer 1996; Anagnostopoulou 2003; Starke 2006), even in the absence of a morphological reflex of these distinctions.

This conclusion is largely based on the behaviour of German stative passives. Consider (19), taken from Salzmann and Schaden (2019, 2):

- (19) Der Computer ist (*von Maria) (*vor drei Tagen) repariert.
 The computer is by Maria before three days repaired.
 ‘The computer is repaired (*by Maria) (*three days ago).’

Assume that Maria repaired the computer three days ago. It still remains the case that neither the agent nor the time of the repair can be stated while using a stative passive. In other languages, such as Greek, stative passives do allow for agents and localizing temporal modifiers that specify the time of the event denoted by the PaPa. Examples such as (19) have commonly been interpreted to indicate that stative passives lack a VoiceP (and by this token, the impossibility of adding an agent), nor an AspP (and thus, the impossibility of specifying the time of the event).

Furthermore, German stative passives are also subject to rather idiosyncratic restrictions with respect to the modifiers they allow (see (20), taken from Gehrke 2015, 904):

- (20) Der Brief ist mit {*dem | einem} Bleistift geschrieben.
 the letter is with the | a pencil written.
 ‘The letter is written with (*the/a) pencil.’

²⁵Deponent *-to* past participles in periphrastic perfects (where they appear with the auxiliary *esse* ‘be’) could be agentive (and thus verbal, not [purely] adjectival) from at least Classical Latin onwards. See also Grestenberger (2018, 499), who cites examples with synthetic (finite) deponents co-occurring with agent-oriented adverbs from a number of older Indo-European languages, including Latin.

Examples like (20) have been argued to show that eventive statives only allow reference to event-type modifiers, but not event-token modifiers (that is, having been written with *a* pencil can be construed as a *type/kind*²⁶ of writing, but not writing with some specific pencil). Furthermore, these modifiers (such as ‘by a child’ in (21)) are not eligible for anaphoric uptake by means of a pronoun. Rather, they are discourse-opaque, which is generally interpreted as a characteristic sign of (pseudo-)incorporation (see Gehrke 2015, 904):

- (21) Die Zeichnung ist von [einem Kind]₁ angefertigt. #Es₁ hat rote Haare.
 The drawing is by a child produced. it has red hair.

Salzmann and Schaden (2019) showed that the PaPa in the German double compound perfect has properties that show more evidence of verbal functional structure than what is assumed for the German stative passive, but less structure than what is assumed for the Greek stative passive. In what follows, we will make a similar argument for the PaPa in Latin HAVE-statives.

2.2.1 Evidence for a Voice-layer?

In the literature on verbal and stative passives and the structure of participles these constructions contain, a general distinction is made between purely adjectival participles, which appear to radically lack an agent participant, and more verbal participles, which do allow for an agent to be present. In terms of structure, agents are assumed to be added by VoiceP, and by this token, diagnosing agentivity is a way of testing whether this functional layer is present. First, note that there are HAVE-statives in which an agent is expressed in a *by*-phrase, as in (12). Another common test is the admissibility of agent-oriented adverbs like *diligenter* ‘carefully’, *libenter* ‘gladly’, *fortiter* ‘strongly’ and *sapienter* ‘wisely’. The idea is that such adverbs cannot combine with states (e.g., one cannot be carefully drunk, or gladly tired). Thus, modification of *habere* can be excluded, and the adverb has to modify the participle. Therefore, if such adverbs are grammatical, this shows us that the structure must contain (at least) a VoiceP.

Agent-oriented adverbs are indeed attested with passive participles in HAVE-statives, as is shown in (22) (in which the accusative subject of the stative small clause is not expressed overtly) and (23).

- (22) tunc [↑]diligenter[↑] *tusos* et *cretos* habebis
 then carefully crushed.ACC.M.PL and sifted.ACC.M.PL have.FUT.2SG
 ‘Then you will have them [grapes] carefully crushed and sifted.’ (Palladius, *De agricultura* 11.14.5, ca. 350 CE)
- (23) scripture, que *diem* et *annum*
 writings.NOM(.F).PL REL.NOM.F.PL day.ACC(.M).SG and year.ACC(.M).SG
habuerint [↑]euidenter[↑] *expressum*
 have.FUTPRF(?).3PL clearly expressed.ACC(.M?).SG

²⁶The literature in syntax like Vergnaud and Zubizarreta (1992) mainly seems to use the distinctions between *types* and *tokens*, whereas the semantics literature after Carlson (1977) mostly talks about *kinds*. We will treat these notions as being interchangeable.

‘writings, which have the day and year clearly mentioned’ (*Lex Visigothorum edita a Reccessvindo* ed. Zeumer, p. 106 (= *Monumenta Germaniae Historica, Leges nationum Germanicarum* vol. 1, 2.5.1), 654 CE)

From examples like (22)–(23) we conclude that PaPas in Latin HAVE-statives may contain at least a VoiceP, and that the relevant past participles are thus more verbal in nature than those appearing, in e.g., German stative passives (see, among others, Rapp 1996).

2.2.2 Reference to event-tokens or only event-types?

The recent literature on (stative) passives has insisted on the distinction between reference to event-tokens and event-types, and it has been established that in German stative *sein*-passives and *haben*-passives, the participle does not denote an event-token, but only an event-type (see, e.g., Gehrke 2012). Once again, the idea is that event-type reference corresponds to a more adjectival PaPa, whereas event-token reference corresponds to a more verbal PaPa. As the distinction between type and token reference of eventive PaPas is best studied for German, we will use examples from this language to illustrate the phenomenon.

As argued in Gehrke (2017), a good criterion to diagnose type-reference is anaphoric uptake with equivalents of *so*:

- (24) a. Maria hat wie wild getanzt, und Jan hat auch so getanzt.
 Maria has like wild danced and Jan has also so danced
 ‘Maria danced wildly, and Jan did so too (lit. has danced in the same way).’
 b. *Maria hat am Dienstag getanzt, und Jan hat auch so getanzt.
 Maria has at.the Tuesday danced and Jan has also so danced
 (intended) ‘Maria has danced on Tuesday, and Jan did so too.’

(24b) is unacceptable under a reading that ‘Jan danced on Tuesday as well’ because the first clause provides only an event-token, but *so* needs to refer to an event-type (which is provided in the first clause of (24a)). Furthermore, different types of event modifiers are compatible with either types or tokens: manner modifiers modify event-types, whereas instruments modify event-tokens.

This test is rather difficult to apply to data from a dead language like Latin, but some relevant evidence is available. In cases like (25), a type-reading seems the only one available (cf. the *sicut*-clause [meaning ‘like, such as’]):

- (25) quia non habebant destinatum unum
 because not have.IPFV.3PL designated.ACC.M.SG one.ACC.M.SG
 sacrificiorum locum sicut Iudaei in
 sacrifices.GEN(.N).PL place.ACC as Jews.NOM(.M).PL in
 Hierusalem
 Jerusalem.ACC(.N).SG

‘because they did not have a single place designated for sacrifices, as the Jews in Jerusalem did’ (Augustine, *De diuersis quaestionibus ad Simplicianum* 1.2.19, ca. 390 CE)

On the other hand, sentences like (26) appear to us to indicate that reference to an event-token was possible in Latin HAVE-statives.

- (26) eos [...] *facies* habere *combustas*
 DEM.ACC.M.PL faces.ACC(.F).PL have.PRS.INF scorched.ACC.F.PL
 eo igne, quem sibi
 DEM.ABL.M.SG fire.ABL(.M).SG which.ACC.M.SG REFL.DAT.M.PL
 succenderant
 light.PLPRF.3PL
 ‘that their faces are scorched by that fire, which they had lit for themselves’
 (Jerome, *Commentarii in Isaiam* 6.13.6, 408 CE)

The PaPa *combustas* ‘scorched’ in (26) appears with the instrumental modifier (*eo igne*, ‘by that fire’). The instrument is determined by a demonstrative, which is not compatible with a stative causal interpretation as described in Maienborn and Herdtfelder (2017). Furthermore, the faces in question seem to be referential, and the interpretation of the events episodic, rather than generic. All this points into the direction of a reference to an event-token.

In other cases it is difficult to disentangle different types of modification. For example, in (27) we have a temporal (*in isdem diebus, qua ...* ‘in the same days as’) and a localizing (*ibi* ‘there’) adjunct, and both of them can specify the state (which has to have token reference). However, spatial localization also applies to the event-token (by entailment), and temporal localization could also shift to the event.

- (27) quia in isdem diebus, qua sanctus
 that in same.ABL days.ABL(.M).PL REL.ABL.F.SG saint.NOM.M.SG
 moyses uel filii israhel contra
 Moses.NOM(.M).SG PRT sons.NOM(.M).PL Israel.GEN(.N).SG against
 illas ciuitates pugnaverant, castra ibi
 DEM.ACC.F.PL cities.ACC(.F).PL fight.PLPRF.3PL camp.ACC(.N).PL there
 fixa habuissent
 fixed.ACC.N.PL have.PLPRF.SBJV.3PL
 ‘that in the same days in which Saint Moses and the sons of Israel fought against those cities, they had struck their camp in that place’ (*Itinerarium Egeriae* 12, ca. 385 CE)

While the evidence is not completely unequivocal, it seems to us that the PaPas in these constructions could be event-token denoting, and not only event-type denoting.

2.2.3 Are event modifiers incorporated?

The last question we will address is whether event modifiers (that is, modifiers of the event argument of the PaPa) in Latin HAVE-statives have to be incorporated. Let us

go back to (26), and the causal modifier *eo igne* ‘by this fire’. As we noticed above, this modifier is not plausibly read as modifying the state, and it also carries both a demonstrative determiner and a relative clause invoking an episodic event where that particular fire had been lit. It seems to us that this indicates that the fire mentioned here must be a token fire, not a fire-type, and that therefore, it is improbable to assume pseudo-incorporation on semantic grounds. Therefore, assuming obligatory incorporation of event modifiers in Latin HAVE-statives seems unconvincing to us.

Let us then briefly sum up section 2.2. Most importantly, the PaPa in the Latin construction seems much more verbal than that in German stative passives: (i) agent-oriented adverbial modification provides evidence for a VoiceP; (ii) token reference for events seems to be possible; and (iii) we found evidence against the obligatory incorporation of modifiers of the event denoted by the PaPa.

2.3 Putting the pieces together

With these results in mind, and building on insights from Maienborn et al. (2016); Alexiadou et al. (2015), we can now proceed to detail the internal structure of a PaPa in a HAVE-stative endowed with the maximal amount of functional structure that we were able to gauge. The derivation in (28) represents a (slightly simplified) analysis of example (4) (the higher part of the functional structure is omitted). At a global level, the top layers of the structure are the same as in (16a), the only difference being the fact that the AP complement of PredP is structurally much more complex than the simple adjective *odiosos* in (16a). We take the participle to be adjectival (which is the contribution of the participial morphology), but we also assume that the AP-layer not only dominates a verb and its internal argument, but also some additional verbal functional structure, most notably a VoiceP (see section 2.2). Furthermore, for reasons that will be elaborated in section 4.1, we assume that Latin did not have grammaticalized definiteness: concomitantly (and with some simplification), we represent nominal phrases as plain NPs.²⁷

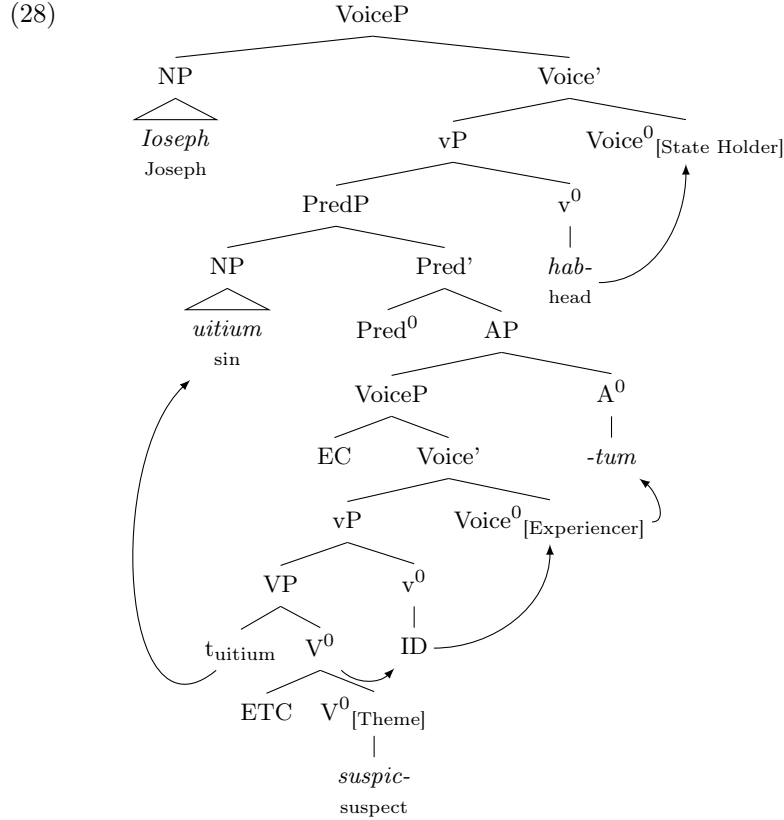
We furthermore assume with Maienborn et al. (2016) that verbs (or verbal roots) start out as purely kind-denoting, and as having kind-type arguments. In order to be token-denoting, and to be able to take token-type arguments, the verb has to undergo *Event Type Closure* (noted as ETC in the tree).²⁸ In Sections 2.2.2 and 2.2.3, we showed that in Latin HAVE-statives, reference to an event-token is possible, and that event modifiers do not need to be incorporated. Therefore, we apply ETC very early. We also assume that *uitium* ‘sin’ receives its θ -role from the participial verb, and that it later moves to the specifier of PredP.²⁹

²⁷Alternatively, the topmost layer of the extended projection of (argument) nouns could be taken to be a KP (for ‘Kase Phrase’, see Lamontagne and Travis 1987; Bittner and Hale 1996)

²⁸See Maienborn et al. (2016, 41) for a thorough description of this operation.

²⁹This analysis is in line with the well-established generalization that the subject of an adjectival passive is thematically the internal argument of the adjectival PaPa, even if in the syntax the subject ends up appearing external to the maximal projection of the predicate (for full discussion, see Levin and Rappaport 1986, and references cited there). Furthermore, as indicated in (28), we take it that at least in this case, Pred⁰ does not assign a thematic role to the element in its specifier. Concretely, with Baker (2003, 36) we assume that whether or not Pred⁰ plays a role in θ -marking depends on the properties of the predicate acting as its complement. More generally, we believe that nothing crucially hinges on our assumption that small clauses contain a PredP; we do not exclude that alternative approaches to small clauses lacking a PredP, as proposed in, for example, Matushansky (2019), may be equally suitable to account for the Latin data under discussion.

We take it that v^0 in the extended projection of the participle is an identity function ‘ID’ (see Myler 2016), that the VoiceP adds the variable for the initiator of the event denoted by the participle, and that the experiencer variable is then existentially closed in the specifier of VoiceP (noted in the tree as EC, for ‘Existential Closure’). The experiencer can in principle be inferred to be the same entity as the subject of HAVE, but this is not a grammatical necessity, but rather a pragmatic effect. In this particular example, it is extremely unlikely that the holder of the state of having suspected a sin is not the one doing the suspecting, and therefore, we get an Attained State reading. With Maienborn et al. (2016), we assume that a partially contextually determined *post-state* – that, is, some state that comes into being when the event reaches its end point – is introduced by A^0 , rather than by Asp^0 (see also Danckaert and Schaden To appear, § 5).³⁰



³⁰Embick (2000) offers an alternative analysis of the Latin PaPa lacking a category-changing adjectival projection: instead, the participial morphology sits in Asp^0 .

2.4 Summing up: HAVE-statives as passive small-clause constructions

Drawing together the main points, we see no need to assume that the complementation behaviour of *habere* is itself variable; any differences between small clauses under *habere* with a plain adjective or with a PaPa as the predicate are due to the amount of functional structure the relevant elements are endowed with. For this reason, we believe that the label of ‘HAVE-statives’ is entirely appropriate to refer to those cases in which *habere* is complemented by a small clause. HAVE-statives with a PaPa predicate with a verbal core simply constitute a special case. It is this subclass of HAVE-statives which in Napoli (2007, 6–10) was identified as that of the *resultativo possessivo* (‘resultative possessive’).

The interpretation of HAVE-statives is characterized by a ‘resultative flavour’, which arises due to the presence of a state argument introduced by the adjectival head transforming the verbal lower structure into a participle. With Maienborn et al. (2016, 354), we take this state to be a contextually shaped result of the event provided by the verbal complex within the participle. In this way, though it is derived differently, it will be rather similar in meaning to a *perfect state* (see, e.g. Nishiyama and Koenig 2010).³¹ In addition, at the level of VoiceP the structure manipulates the argument structure of the participle, where the external argument of *habere* is not grammatically identified with the external argument of the verbal complex, giving the pattern a ‘passive flavour’.

With our analysis of Latin HAVE-statives (with a verbal PaPa) in place, we can consider how and why this pattern could develop into a true perfect tense. We will argue that grammaticalized definiteness in inalienable structures played a crucial role for this development. In section 3 we will spell out our assumptions about the syntax of inalienable possession constructions featuring a definite article. In section 4 we then spell out our assumptions about the syntax of Latin noun phrases (with special reference to those appearing in inalienable possession environments), before putting all the pieces together in section 5.

3 Setting the scene: definite articles and inalienability

In this and the following sections, we turn to the developments that ultimately caused the genesis of the Romance periphrastic HAVE-perfect. As mentioned, we will explore the hypothesis that the key catalyst was an independent change, namely the grammaticalization of a D-layer, as expressed by means of definite articles. We start by giving some background information on definiteness and inalienability, after which we turn to the diachrony of definite articles in the Romance language family.

3.1 Inalienable possession and syntactic binding

Inalienable possession can be defined as an asymmetric relation between two entities X and Y, to the effect that Y is characterized as an inherent or typical attribute of X. Arguably the most prototypical type of inalienable possession is the part-whole

³¹For a full compositional analysis of the construction, see Danckaert and Schaden (To appear, § 5).

relation, as instantiated, for example, by humans or animals and their body parts. Many languages employ one or more specialized morphosyntactic strategies to encode inalienable possession (see, among many others, the contributions to Chappell and McGregor 1995), to the effect that inalienable possession is marked differently (and typically less explicitly) than alienable possession, the latter being defined as an asymmetric association between two entities which is more *ad hoc*, less permanent in nature. As pointed out in Rooryck (2022), the term ‘inalienable’ is not to be interpreted literally: for example, items of clothing may enter into syntactic configurations of inalienable possession, but there is of course no sense in which the relevant items can literally not be taken away from their possessors. Rather, the set of elements that may be treated on a par with true inalienable parts can be said to contain “nouns that are commonly expected to be located in or on the possessor” (Rooryck 2022, 1). Nevertheless, in what follows we will continue to use the traditional label ‘inalienable’.

The upcoming discussion is largely based on the classic analysis of inalienable possession constructions in present-day French from Vergnaud and Zubizarreta (1992). We will primarily focus on French, but the core of the analysis carries over to other Romance languages. Furthermore, we are only concerned with structures in which an inalienably possessed nominal acts as the internal argument of a transitive predicate, which, as we will claim below, is the crucial syntactic environment relevant for the development of HAVE-perfects.

3.1.1 Two types of inalienably possessed objects

In present-day French, there are two types of construal for inalienable direct objects, one with and one without a dative argument acting as the external possessor (a useful overview of the relevant empirical data can be found in Junker and Martineau 1987).³² The basic patterns are illustrated in (29):

- (29) a. Les élèves lèvent [ʁla[⌞] main].
the.PL pupils raise.PRS.3PL the.SG hand
‘The pupils raise their hand.’
b. Les élèves *(se) lavent [ʁle[⌞] visage].
the.PL pupils REFL wash.PRS.3PL the.SG face
‘The pupils wash their face.’
c. Les élèves *(leur) lavent [ʁle[⌞] visage].
the.PL pupils DAT.3PL wash.PRS.3PL the.SG face
‘The pupils wash their face.’

In the case of verbs like *lever* ‘lift’ (29a), the external possessor of the inalienably possessed direct object is the agent/subject of the clause, whereas with verbs like *laver* ‘wash’ (29b)–(29c), the external possessor is realized as a dative. The external possessor may or may not corefer with the subject. When such coreference obtains, the dative takes the shape of a reflexive pronoun *se* (29b); otherwise it is realized as a non-reflexive dative pronoun (like *leur* in (29c)) or as a PP headed by *à* ‘to’ (see e.g.

³²For further discussion of the French data, as well as a comparative perspective on Spanish and Portuguese, see Spanoghe (1995). For a more recent discussion of the French data and their analysis, see Rooryck (2022).

example (31c) below).³³ Second, as noted in Hatcher (1944) the difference between the two constructions is correlated with a semantic difference: *lever*-type verbs typically express an action *performed with* a body-part, whereas *laver* and similar verbs tend to describe an action *done to* a body-part (see also Herschensohn (1975), who refers to the former type as ‘pseudo-transitives’). Third, with respect to the diachronic origins of the two types of inalienable object constructions, according to Stéfani (1962, 318–320) the pattern in (29a) is older than that in (29b). For grooming verbs like *laver* ‘wash’ and other verbs belonging to the same class, Old French typically did not use a reflexive dative external possessor, but rather an adnominal possessive pronoun or, more rarely, a determinerless relational noun. Stéfani further suggests that the diachronic primacy of these two alternative strategies is to be expected, given that they are both inherited from Latin.³⁴ On the other hand, in contrast with (29b), the pattern in (29c) does seem to be a more direct continuation of Latin. Late Latin examples of non-reflexive dative arguments acting as the external possessor of an inalienably possessed direct object are illustrated below in (47) and (41a)–(41b) (see also Gianollo 2014), but the same construal is also attested in earlier stages of the language (Baldi and Nuti 2010, 350–355), as well as elsewhere in older Indo-European languages (for a comparative overview, see Havers 1911; in particular on Latin, cf. pp. 170–239). It is conceivable that the pattern in (29b) came into being through analogy with that in (29c) (although this is difficult to prove).

The examples in (29) are also interesting with respect to the usage of the definite article. In each case, under the intended reading, the definite article has to take narrow scope with respect to a distributive operator (see, e.g., Champollion 2017). Second, at least in (29a), the definite article does not seem to come with its habitual uniqueness presupposition: (29a) does not require the pupils in the class to be all one-armed, which we would expect if for every pupil, there is a unique hand they could raise, nor does it necessarily entail that they raised only one hand, instead of both. The consensus view seems to be that the meaning of the definite article in this context is reduced compared to its ‘standard’ meaning.³⁵

³³The asterisks in (29b)–(29c) indicate that the dative argument is obligatory in order for the inalienable reading to obtain; if the dative is omitted, the sentence is grammatical, but the object *le visage* ‘the face’ cannot receive a distributive interpretation.

³⁴Thus Stéfani (1962, 320): “En résumé, là où, aujourd’hui, on rencontre, le plus souvent l’accusatif de relation avec le verbe pronominal, on employait plutôt, en ancien français, le tour : « nominatif de la partie du corps + datif du pronom représentant la personne + verbe actif », s’il s’agit d’un accident ou d’un incident survenu au héros, et le tour, « verbe transitif + *son* (*le*) + nom de la partie du corps », pour une activité normalement exercée par l’individu sur lui-même, cette seconde tournure pouvant d’ailleurs s’employer aussi dans le premier cas. Les deux constructions sont « étymologiques ». En latin, on disait normalement: *pedes lavat* (*pedes suos lavat* et *suos pedes lavat* traduisant des intentions stylistiques particulières) et *oculi mihi dolent*. Au contraire, *sibi lavare manus* est inconnu de la langue classique. C’est une création romane ou française.” (“To sum up, in those cases where today one most often finds a relational accusative [*a relational noun with a definite article*] with a pronominal verb, in Old French, one would rather use the pattern “nominative of a body part noun + dative pronoun corresponding to the person + active verb”, used when something happened to the protagonist, whereas the pattern “transitive verb + *son* (*le*) [*his* (*the*)] + body part noun” was typically used for actions performed by an individual to themselves (although this structure could also be used in the previous case). These two constructions are “etymological”. In Latin, one usually said *pedes lauat* [*feet washes*] (with the variants *pedes suos lauat* [*feet his washes*] and *suos pedes lauat* [*his feet washes*] conveying particular stylistic connotations) and *oculi mihi dolent* [eyes me.DAT hurt]. In contrast, *sibi lavare manus* [*oneself.DAT wash hands*] is not known in the classical language. It is a Romance or French innovation.”)

³⁵What exactly this reduced meaning is, is not that important; Vergnaud and Zubizarreta (1992) suggest that it has no meaning at all, whereas Rooryck (2022) assumes that they are weak definites. Notice, however,

With this background information in place, we now turn to the syntactic relation between an inalienably possessed object and its external possessor. Specifically, we will review the evidence put forward in Vergnaud and Zubizarreta (1992) in favour of the claim that these two elements should sit in a local structural configuration.

3.1.2 The structural position of the external possessor

At the heart of the proposal by Vergnaud and Zubizarreta (1992) is the idea that French definite articles come in a meaningful and an expletive (semantically vacuous) version,³⁶ which are homophonous but can be distinguished on syntactic and semantic grounds. First of all, it is important to note that (29a) is ambiguous between an alienable and an inalienable reading. The latter is arguably the more prominent one: its signature property is that a singular inalienably possessed object associated with a plural external possessor gives rise to a distributive reading. In the case at hand, as many hands are being raised as there are pupils. This effect is absent in the – pragmatically less likely – alienable reading, under which the pupils raise one single, disembodied hand. This ambiguity disappears when the DP acting as the direct object is headed by a non-relational noun. For example, (30) is only compatible with an interpretation which involves a single piano (irrespective of whether one or more lifting events are being referred to). In contrast, the pattern in (29b) must feature a relational noun. Relatedly, (29b) and (29c) never display the same ambiguity that is found with *lever*-type verbs, as the inalienable – and in this case also distributive – reading is the only one available.

- (30) Les élèves lèvent [ʁle piano].
 the.PL pupils lift.PRS.3PL the.SG piano
 ‘The pupils lift the piano.’

As argued for in detail in Vergnaud and Zubizarreta (1992), the distributive reading can only obtain if the possessor argument of the relational noun is bound by a suitable antecedent. Importantly, this binding relation needs to be established syntactically. In addition, strict locality conditions are imposed on the relation between the possessor and the possessee.³⁷ To illustrate the relevant conditions, consider the following data from Vergnaud and Zubizarreta (1992, 620) (the examples are their (57a,b); the asterisks indicate that the examples are ungrammatical under the desired inalienable reading). In (31a), the possessor DP *Pierre* and the inalienable object *la tête* ‘the head’ are located in two separate coordinated clauses. As a result, the external possessor does not c-command the possessee, and no bound reading can obtain. In addition, a mere c-command relation is not sufficient: in (31b), the matrix subject *Pierre* does in fact c-command the inalienable noun, but the two phrases are not

that these are both rather ‘bleached’ versions of a definite meaning, and far removed from a demonstrative meaning.

³⁶Again, whether the definite article is truly vacuous or merely semantically reduced (as suggested by Rooryck 2022) does not really change the nature of the argument we are making here. Note also that the term ‘expletive article’ is used in a slightly different way in Vergnaud and Zubizarreta (1992) and in Longobardi (1994, 646–652): in the latter study, it refers to a definite determiner used in some languages with proper names and kind-referring nouns.

³⁷The exact nature of the locality conditions imposed on French inalienable possessions constructions has been discussed since Kayne (1975) (see also Guéron 1985; Authier 1988).

located in the same clause: in this non-local configuration, no binding relation can be established. Finally, in the case of *laver*-type verbs, a subject external possessor can never act as the antecedent of an inalienable possessee in the presence of a dative (it can only do so indirectly, when the dative is a reflexive pronoun). This is illustrated in (31c), where *les mains* ‘the hands’ belong to Dominique, not to Pierre. This indicates that apart from the c-command and clausemate conditions established on the basis of (31a)-(31b), the binding category must also be the closest c-commanding DP.

- (31) a. *Pierre est tombé et [l’a¹ tête] a heurté le
 Pierre be.PRS.3SG fallen and the.SG head have.PRS.3SG hit the
 trottoir.
 sidewalk
 (intended) ‘Pierre fell and hit his head on the sidewalk.’
 b. *Pierre pense que [l’e¹ bras] est cassé.
 Pierre think.PRS.3SG that the.SG arm be.PRS.3SG broken
 (intended) ‘Pierre thinks that his arm is broken.’
 c. Pierre lave [l’es¹ mains] à Dominique.
 Pierre wash.PRS.3SG the.PL hand.PL to Dominique.
 ‘Pierre washes Dominique’s hands.’

How can the locality constraints on the French inalienable possession construction be stated more formally? Vergnaud and Zubizarreta (1992) propose that the possessor and the possessee have to mutually m-command each other.³⁸ As the notion of m-command is now largely obsolete, here we will reformulate the relevant locality constraint in more general terms, by saying that the external possessor is the closest DP which asymmetrically c-commands the possessee, and that no clause boundary may intervene between the two.³⁹ In cases like (29a), which do not feature a dative argument, the subject presumably binds the possessee under reconstruction, that is, at the stage of the derivation where it sits in its base position SpecVoiceP.⁴⁰

Though we will adopt the framework from Vergnaud and Zubizarreta (1992) to detail our analysis of the grammaticalization of auxiliary HAVE, we are aware that

³⁸The relation of m-command is slightly broader than that of c-command. Specifically, a node X m-commands (i) all the nodes it c-commands, (ii) as well as any node that c-commands X and which is itself dominated by the maximal projection of X.

³⁹These locality constraints are reminiscent of Principle A of the Binding Theory, which governs anaphor binding: see Rooryck (2022, 21) for a unified syntax for (what are traditionally called) inalienable possession constructions and binding of (simplex) reflexives.

⁴⁰Two types of alternative analyses have been proposed to bring these structures in line with those in which a dative antecedent is obligatory (cf. (29b)). According to Guéron (1985, 62) (followed by Vergnaud and Zubizarreta 1992, 621–622), in the relevant cases the verb together with its internal argument is reanalysed as a complex predicate: “[w]ithout changing the syntactic structure, reanalysis triggers absorption of the theta-role of the object by the verb. The NP loses its independent status and may no longer be bound.” An additional interpretive rule is invoked to ensure that the inalienable reading can be derived. There are also semantic reasons to treat cases like (7b) as non-transparent idioms, contrary to cases like (29b). Singular definite determiners are generally assumed to carry (situational) presuppositions of uniqueness. For instance, if I say, “I saw the teacher”, there needs to be a contextually unique teacher, identifiable by the hearer, in order for the sentence to be felicitous. Notice that this is not the case for sentences like (7b): this sentence does not suggest that the pupils are all amputees that have only one remaining hand. Rather, somehow the uniqueness presupposition of the definite article gets neutralized. Yet another proposal, more in line with the approach adopted here, is developed in Herschensohn (1996), who argues that the external argument in examples like (7a) is not in fact a thematic agent, but rather a derived subject which originates as an indirect object bearing the thematic role of possessor. Binding of the inalienable theme argument happens locally when the possessor sits in its VP-internal base position (that is, under reconstruction).

this is neither the only nor the most recent proposal available. One recent alternative is Rooryck (2022). Focusing primarily on data from French, the author proposes to radically dissociate the traditional categories of alienable and inalienable possession, arguing that the latter has in fact nothing to do with inalienability *stricto sensu*, nor with possession. Instead, he argues that the family of constructions traditionally subsumed under the label ‘inalienable possession’ (which in Romance feature the trademark definite determiner) involves a locative relation between two entities, which is encoded in a structure like (32) (adapted from Rooryck (2022, 21), his (55b)), where ‘R(P)’ stands for ‘Relator (Phrase)’ (in the sense of den Dikken 2006). In this structure, the location (‘possessor’) originates in a structurally lower position than the locatum (‘possessee’), but this order is reversed when the former A-moves across the latter to SpecTP:

$$(32) \quad [_{TP} \text{ LOCATION } [_{T'} \text{ HAVE } (= \text{BE} + \text{R} + \text{P}) \quad [_{RP} \text{ LOCATUM } \text{R} + \text{P} \quad [_{PP} \text{ P}_{\text{locative}} \text{ LOCATION}]]]]]$$

Two of the main differences between this analysis and that of Vergnaud and Zubizarreta (1992) are the following. First, according to Rooryck (2022), the lexical noun heading the phrase which expresses the location is not relational (that is, it is not subcategorized for a possessor argument). Concomitantly, there is no sense in which the entire construction expresses possession. Rather, the close relation between locatum and location derives from the lexical semantics of the locative preposition. Second, from a semantic point of view, the DP headed by what Vergnaud and Zubizarreta (1992) consider an expletive article is taken to be a weak definite (in the sense of Schwarz (2009); Aguilar-Guevara (2014), among others).

Despite these important differences, two properties of the French inalienable possession constructions we have reviewed above and which we consider crucial for the purpose of the upcoming diachronic analysis can be accounted for under both analyses. First, for Vergnaud and Zubizarreta (1992), the inalienably possessed nominal necessarily needs a binding category. Although there is no binding relation in the analysis of Rooryck (2022), a very similar effect derives from the argument structure of the locative preposition: the latter is a two-place predicate, which needs both a complement (the location) and a subject (the locatum). Second, the fact that it needs a *local* binder (cf. the examples in (31)) can straightforwardly be made to follow from familiar locality constraints on A-movement.

In the upcoming diachronic analysis, we will use the formalism of Vergnaud and Zubizarreta (1992), but any theoretical framework that can account for the data in (31) would be suitable.

3.2 Type reference and expletive determiners

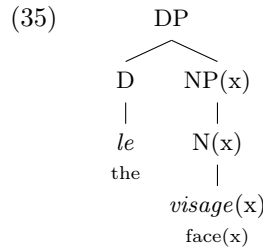
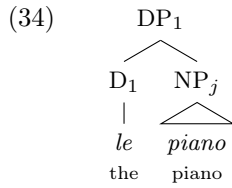
We now turn to the interpretive properties of inalienable possession constructions, and to the way these are encoded in the syntax. Vergnaud and Zubizarreta (1992) propose that in cases like (33a) (= (30)), reference is made to a token (of a given entity), whereas in (33b) (= (29b)), we are dealing with type reference.⁴¹

⁴¹ Assuming that the definite DP is a weak definite, and analyzing it as kind-denoting is one of the main formal accounts (for a recent overview, see, e.g., Aguilar-Guevara and Oggiani 2023).

- (33) a. Les élèves lèvent [¹le¹ piano].
the.PL pupils raise.PRS.3PL the.SG piano
‘The pupils lift the piano.’
b. Les élèves *(se) lavent [¹le¹ visage].
the.PL pupils REFL wash.PRS.3PL the.SG face
‘The pupils wash their face.’

In terms of syntactic structures, the authors assume that token reference is expressed at the level of D(P), and type reference at the level of N(P). Vergnaud and Zubizarreta (1992, 612) dub this generalization the *Correspondence Law*: “[w]hen a DP or an NP denotes, the DP denotes a token and the NP denotes a type”.

To make this more concrete, let us have a look at the syntax of the two construals exemplified in (33). The structure of a token-denoting DP with a contentful article is as in (34) (the two following trees were adapted from Vergnaud and Zubizarreta 1992, 613, 615):



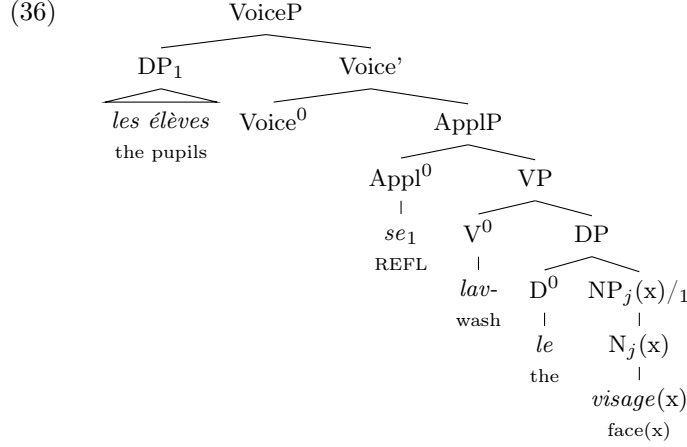
In (34), two indices are represented, the lower one ‘j’ signaling type reference, and a higher index ‘1’ indicating token reference. The complementation relation between D and NP is taken to mediate between those two types of reference, to the effect that the addition of a semantically active determiner ensures that a given type is *instantiated* by a particular token, despite the fact that the NP itself denotes a type.

On the other hand, Vergnaud and Zubizarreta (1992) assume the structure of the direct object in (33b) to be a DP with an expletive article: the latter is added for reasons of syntactic well-formedness (in French, a nominal argument canonically takes the shape of a DP), but is semantically completely inert.⁴² The structure of a DP with an expletive article is given in (35).

In (35), the noun *visage* ‘face’ is to be interpreted as inalienably possessed, and as a consequence, its reference is determined by an element higher up in the clause. Therefore, we cannot directly attribute an index to it here. Vergnaud and Zubizarreta (1992) represent the possessor by the variable *x*, which needs to be bound in order for reference to a type to be established. According to Vergnaud and Zubizarreta (1992, 608–609), type reference of NP in a DP like (35) comes about through local binding by a (token-denoting) DP, which qualifies as the external possessor of the relational noun. A simplified structure for (33b) would thus look like (36). In this structure, binding of

⁴²Romance languages other than French are more liberal in allowing for determinerless nominal arguments, but underlyingly, these may well be DPs: see section 5.1.2. We remain agnostic as to why across Romance, inalienable possession constructions feature an expletive (weak) definite determiner, rather than, e.g. DPs with a covert determiner.

the variable index of N by the dative reflexive pronoun *se* (syntactically a clitic, represented here as the head of an applicative phrase) gives rise to the desired distributive reading, under which a list of pairs of pupils and matching faces is established.⁴³



As mentioned in section 1.2, Latin did not have definite articles, which raises the question as to how distributive vs. non-distributive readings of nominals, and relatedly, the contrast between alienably and inalienably possessed NPs, could be expressed in the language. We address this topic in section 4.

4 Expressing inalienable possession in Latin

4.1 Latin as an NP language

The starting point of this section is the observation that Latin did not have definite articles, nor any other morphosyntactic strategy that was narrowly specialized to express definiteness. We propose that the absence of grammaticalized definiteness in Latin implies that the language did not have a D-projection. To use the terminology from Bošković (2008, 2009) (and related literature), we may say that Latin was an NP-language, in which the morphosyntactic category D was radically absent. Concomitantly, all adnominal determiners that are inherently definite, such as demonstratives, are merged in specialized functional categories different from D.

Some authors have argued that despite the absence of an article system, Latin did in fact have a DP-layer (see in particular Ledgeway 2012 and Giusti and Iovino 2016). For one thing, it is no doubt true that Latin had complex nominal groups, such as the one illustrated in (37):

- (37) illud meum turbulentissimum tempus
 DEM.NOM.N.SG POSS.NOM.N.SG turbulent.SUP.NOM time.NOM(.N).SG

⁴³In the more recent literature (cf., e.g., Champollion 2017), the mechanism to achieve distributivity is implemented differently, for instance via an applicative head D(istributive). This does however not change the – for our purposes, crucial – necessity for the possessor argument to be bound by a higher NP or DP in such contexts. Since the presence of such a distributive head is not essential for our argument, in the remainder of the paper, we will omit it.

‘this most turbulent time of mine’ (Cicero, *In Pisonem* 33, ca. 55 BCE)

This example features three adnominal modifiers, namely a demonstrative, a possessive and an adjective, each presumably hosted in the specifier of a dedicated functional projection. The category of demonstratives is of special interest, because it is arguably the best candidate to fulfil the function of ‘determiner’. However, the availability of adnominal demonstratives can hardly be construed as an argument in favour of a DP-analysis of Latin. For one thing, all natural languages appear to have at least one type of demonstrative (Diessel 1999), and with vanishingly few exceptions (cf. Dixon 2003, 69), they all have nominal demonstratives, that is, demonstratives that are either used adnominally or purely pronominally. As noted in Dixon (2003, 65), the adnominal usage is more widespread than the pronominal one, as no language appears to have the latter but not the former (whereas the reverse distribution is in fact attested). In other words, if the existence of adnominal demonstratives were to be used as a diagnostic for DP-hood, then the whole NP/DP debate would become moot. More generally, although ‘D’ in DP stands for ‘Determiner’, the NP/DP debate is not about whether a language has (adnominal) determiners, it is about whether it has (definite) articles (which are, by assumption, hosted in the functional projection D), and if so, if it harnesses them to morphosyntactically encode an asymmetry between nominal arguments and nominal predicates (at least in some syntactic contexts): see Longobardi (1994) and Chierchia (1998).

Furthermore, Ledgeway (2012, 261–269, 281–282) suggests that the presence of a D-projection can be diagnosed indirectly, by looking at the word order behaviour of adnominal modifiers and complements. The example in (38) from Caesar, cited in Ledgeway (2012, 282), is a good case in point (bracketing and labels slightly modified from the original):

- (38) [DP [AP celeris] [D' [NP [N' spe [DP t_{celeris}
swift.GEN.N.SG hope.ABL(.F).SG
subsidii]]]]
assistance.ACC(.N).SG
'[encouraged] by the hope of swift assistance' (Caesar, *De bello civili* 3.69.2,
ca. 45 BCE)

In this example, the noun *spe* ‘hope’ takes a nominal complement (*subsidiū* ‘assistance’); the latter is modified by the adjective *celeris* ‘swift’, but this element appears to the left of the head of the entire complex nominal group (*spe*). Given the presence of the intervening element (*spe*), the order ‘adjective - noun’ is clearly not due to absence of N-movement across the AP (where the latter would be first merged in the extended projection of the former). Rather, with Ledgeway (2012) we may assume that the adjective *celeris* has moved to a position at the edge of the complex nominal group, presumably for reasons related to information structure. As indicated in (38), Ledgeway assumes that this edge position corresponds to SpecDP, which he furthermore takes to be the edge of a phasal domain (in the sense of Phase Theory, see Chomsky 2001 and subsequent work).

An even more forceful argument to the same effect can be made on the basis of the word order behaviour of adnominal *wh*-modifiers, such as the genitives *cuius* ‘whose.SG’ (39a) and *quorum* ‘whose.M/N.PL’ (39b). As pointed out in Danckaert (2012, 152), the otherwise productive variation between the orders ‘genitive - noun’ and ‘noun - genitive’ is suspended in the case of *wh*-genitives, which invariably appear prenominal (in fact, they must not only precede the head noun, but also all other material in a complex nominal group). This effect is observed with relative (39a), interrogative (39b) and exclamative *wh*-words (the latter case is not illustrated here for reasons of space). Word orders like *umbra cuius* and *nomina quorum*, where the *wh*-genitives surface postnominally, are never attested, and given the size of the corpus and the overall token frequency of adnominal *wh*-genitives, these patterns can safely be assumed to have been impossible in Latin.

- (39) a. quercus [. . .] [[⌈]cuius[⌋] umbra
 oak.NOM(.F).SG REL.GEN.F.SG shadow.NOM(.F).SG
 opaca] sedes erat
 thick.NOM.F.SG seat.NOM(.F).SG be.IPFV.3SG
 ‘an oak, whose dense shade offered a resting-place’ (Livy, *Ab urbe condita*
 3.25.7, ca. 10 CE)
- b. [[⌈]Quorum[⌋] nomina] prius parentes
 Q.GEN.M.PL names.ACC(.N).PL earlier parents.NOM(.M).PL
 liberis suis ingerunt?
 children.DAT(.M).PL REFL.POSS.DAT.M.PL force.upon.PRS.3PL
 ‘Whose names do parents force upon their children’s ears earlier?’
 (Tacitus, *Dialogus de oratoribus* 7.4, (shortly) after 100 CE)

A natural way to account for the strictly prenominal placement of Latin *wh*-genitives is to assume that the language had obligatory internal *wh*-movement (for seminal discussion of this phenomenon, see Aissen 1996): that is to say, whenever an item X carrying a *wh*-feature pied-pipes a larger constituent Y to the edge (left periphery) of domain Z, X always first moves to the edge of Y before Y itself moves to the edge of Z.⁴⁴ Similarly, we may assume that whenever a *wh*-item is subextracted (left-branch extracted) from a complex nominal group, the relevant item first moves to the edge of the complex nominal, and then moves on to its final landing site.⁴⁵ Note in passing that it remains to be seen what the exact nature is of the edge position of NP: according to Cinque (2014), phrasal movement out of DPs in Italian proceeds via an A position (rather than an A-bar position); given our current state of knowledge, it is unclear whether this is the case for extraction out of Latin NPs too.

Most importantly, there is no reason to believe that the existence of a series of functional projections within the nominal domain entails the existence of a D-layer,

⁴⁴As discussed in Danckaert (2012, 152–153, fn. 15), in Latin the one exception to this generalization is found in the prepositional domain: in PPs, an adnominal *wh*-item either precedes or directly follows the preposition.

⁴⁵With respect to discontinuous nominal groups in which the lefthand element does not bear *wh*-morphology, Ledgeway (2012, 264–265) proposes that the same edge position acts as an escape hatch for subextracted phrasal material, yielding the characteristic discontinuous nominal groups that Latin is famous for. However, according to Bošković (2008, 2009), left branch extractions of adnominal modifiers are typical of NP-languages, not of DP-languages.

- c. ut [...] *pro* inclinent [caput
 so.that bow.PRS.SBJV.3PL head.ACC(.N).SG
 suum] sacerdotibus
 REFL.POSS.ACC.SG priests.DAT(.M).PL
 ‘so that they bow their head to the priests’ (Rufinus (translator of
 Origenes), *In Iesu Naue homiliae XXVI* 10.3, 403–404 CE)

All three examples feature a transitive finite verb with a third person plural subject and the bare noun *caput* ‘head’ as the direct object. In (40a), the object NP denotes a token: reference is made to the disembodied head of King Saul. In contrast, a (distributive) type-referring interpretation is the intended one in (40b): the individual members of the class of priests are forbidden to shave their head. Crucially, this interpretation is not accompanied by any morphosyntactic marking: not only is the shape of the direct object identical to that in (40a), it is also not necessary to add a dative (reflexive) possessor in the case of (40b) (which is obligatory with the French verb *raser* ‘shave’). Finally, as shown in (40c), the direct object can also be accompanied by a (reflexive) possessive determiner, again bearing singular morphology.

The situation is slightly different when the external possessor of an inalienable possessee does not correspond to the subject of the clause: in those cases, the external possessor typically takes the shape of a dative-marked NP or pronoun (highlighted in boldface in (41a–b)); note the distributive reading in the latter example), or alternatively as an adnominal genitive (41c) or a possessive determiner. Occasionally, the external possessor can also be left unexpressed (as is the case in (41d)):

- (41) a. Unxit *ei* [caput].
 anoint.PRF.3SG DEM.DAT.M.SG head.ACC(.N).SG
 ‘He anointed his head.’ (Caesarius of Arles, *Sermo* 59.7, ca. 520 CE)
- b. Fouitionibus frigidis magis posca
 poultices.ABL(.F).PL cold.ABL.F.PL rather posca.ABL(.F).SG
eis [caput] fouebis.
 DEM.DAT.N(?) .PL head.ACC(.N).SG foster.FUT.2SG
 ‘You must rather apply cold poultices with *posca* [a mixture of vinegar and water] to their heads.’ (*Mulomedicina Chironis* 59.7, 4th c. CE (?))
- c. quod unxit [[caput] *eius*]
 because anoint.PRF.SBJV.3SG head.ACC(.N).SG DEM.GEN.M.SG
 ‘because she anointed his head’ (Jerome, *Commentarii in euangelium Matthaei* 4.26.14, ca. 400 CE)
- d. At illi magistri sanguinem
 PRT DEM.NOM.M.PL teacher.GEN(.M).SG blood.ACC(.M).SG
 sitientes, ceratas
 be_thirsty.PTCP.NOM.M.PL coated_with_wax.ACC.F.PL
 [caput] inlidunt tabellas.
 head.ACC(.N).SG beat.PRS.3PL tablets.ACC(.F).PL
 ‘Thirsting for the blood of their teacher, they hit <him> on the head with their wax tablets.’ (Gregory of Tours, *Liber in gloria martyrum* 42, ca. 580 CE)

Before we proceed, let us point out that in the case of inalienably possessed direct objects, the use of a singular noun is not the only possible option: rather, plural nouns are also attested in this environment.⁵⁰ The examples in (42) illustrate a number of possible variants:

- (42) a. et inpende in illis ut ꞑproꞑ radant
 and pay.IMP.2SG onto DEM.ABL.M.PL that shave.PRS.SBJV.3PL
 [capita]
 head.ACC(.N).PL
 ‘and give them money so that they can have their heads shaved’ (*Vulgata, Acta apostolorum* 21.24, ca. 405 CE)
- b. ut ꞑproꞑ inclinent [capita sua]
 that bow.PRS.SBJV.3PL head.ACC(.N).PL REFL.POSS.ACC.PL
 omnes
 all.NOM.M.PL
 ‘that all should bow their head’ (*Itinerarium Egeriae* 25, ca. 385 CE)
- c. manus, [capita] ꞑsororibusꞑ osculabatur
 hand.ACC(.F).PL head.ACC(.N).PL sister.DAT(.F).PL kiss.IPFV.3SG
 ‘He kissed his sisters on the hands, on the head.’ (Ambrose of Milan, *De obitu Valentiniani* 36, 392 CE)

(42a) features the bare noun *capita* ‘heads’.⁵¹ In (42b), the same plural noun is modified by the reflexive possessive *sua*. Finally, (42c) is an example in which a plural indirect object (the dative noun *sororibus* ‘sisters’) acts as the possessor of the body part.

At this stage it is not clear to us what governs the distribution of singular and plural nouns in Latin inalienable possession constructions. We certainly do not exclude that there may be subtle semantic or usage-based differences between the variants illustrated in (40)-(42), and/or that the relevant alternations are subject to diachronic variation. However, the mere observation that there is syntactic variation is in line with our contention (to be elaborated shortly) that Latin did not have a specialized syntactic construction to express inalienable possession. In any event, in what follows we will not further be concerned with the pattern with a plural noun, which we do not believe to be directly relevant for the development of periphrastic perfects with HAVE (to our knowledge, structures like (42) are not preserved in the Romance daughter languages).

To sum up the discussion so far, the facts just reviewed lend support to our proposal that the category D is not involved in the expression of inalienable possession relations in Latin, and that the type-referring reading of inalienably possessed nouns does not involve a syntactic binding configuration in which an external possessor acts as the binding category. Rather, we take it that the relevant interpretations are contextually determined.

⁵⁰See also example (26), which features the plural noun *facies* ‘faces’.

⁵¹The fact that in (42a), the intended agent of the shaving event may not coincide with the possessor of the body-part noun is irrelevant: what counts is that the latter corefers with the syntactic subject of *radant* ‘shave (SBJV)’ (this is indeed the case). It is also worth noting that the Greek original of this passage features the singular DP *tên kefalên* ‘the head’.

4.3 Type-referring nominals and HAVE-statives

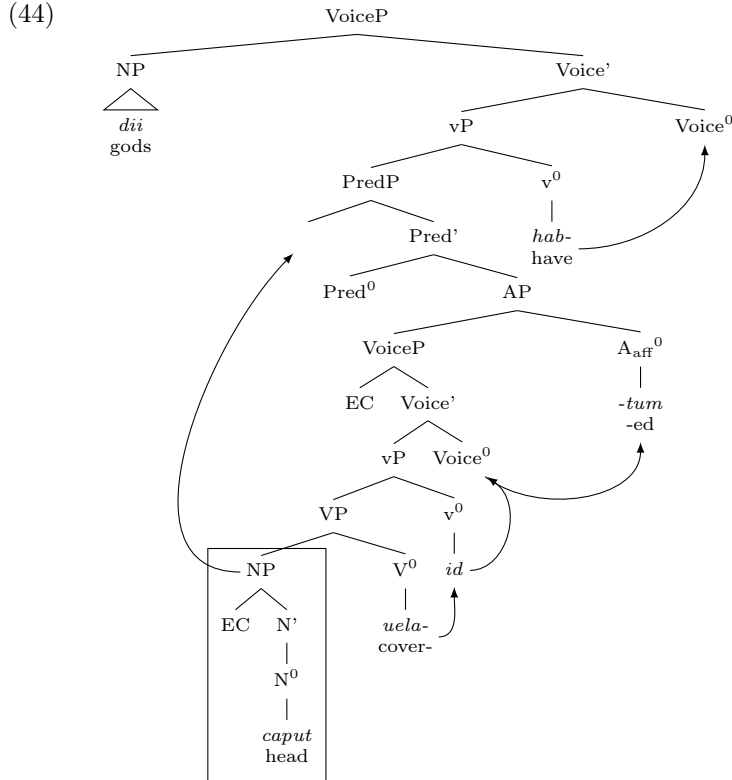
Consider next the following Late Latin example, which contains an external possessor configuration with an inalienable possessee embedded in a HAVE-stative.

- (43) nam [dii qui erant apud
 PRT gods.NOM(.M).PL REL.NOM.M.PL be.IPFV.3PL at
 Laurolauinium] non habebant [*uelatum*
 Laurolavinium.ACC(.N).SG not have.IPFV.3PL veiled.ACC.N.SG
 caput].
 head.ACC.(N).SG
 ‘because the gods who were at Laurolavinium did not have their heads
 covered’ (*Seruius auctus*, *liber* 3, *ad uersum* 174, 6th c. CE(?))

The intended reading of this example is one in which none of the gods who were present at Laurolavinium had their head covered: this distributive interpretation obtains with a singular noun as the subject of the small clause (viz. *caput* ‘head’). Given that Latin did not require an overt adnominal possessive determiner, and as there is no sign of a grammaticalized distinction between alienable and inalienable possession, there can be no binding of the understood possessor of the relational noun *caput* ‘head’ along lines of Vergnaud and Zubizarreta (1992). Instead, we take it that the antecedent of the possessor argument has to be retrieved pragmatically, and that the inalienable interpretation arises if the antecedent is equated with the subject of *habere*.

A possible structure of the relevant portion of this example is given in (44) (we are omitting TAM-related and higher projections on top of the VoiceP projected by the HAVE-verb). As above, the abbreviation ‘EC’ stands for existential closure: note that we assume that both the possessor argument of *caput* ‘head’ and the agent of *uelare* ‘veil’ are existentially closed.⁵²

⁵²The option of representing the possessor argument of the relational noun as existentially closed would not be appropriate for French examples like (29b), as a structure of this kind would overgenerate readings. Concretely, the fact that the distributive (inalienable) reading is the only possible one for examples like (29b) and (29c) would not be accounted for. The crucial point is that the corresponding Latin constructions are as a rule structurally ambiguous, unlike French (29b)–(29c), where inalienable possession is encoded explicitly by means of the dative argument and the (expletive) definite article. It seems plausible to extend this reasoning to other inalienable possession constructions like (29a), which are in principle ambiguous.



In this structure, the (root of the) lexical verb *uelare* ‘cover (with a veil)’ selects an NP complement (namely *caput* ‘head’). Importantly, as indicated, we assume that this nominal category does not have any referential index at all, neither a constant nor a variable. We assume furthermore that the understood possessor argument of the relational noun *caput* is syntactically projected in SpecNP, but that it is immediately existentially closed. In this way, the grammar tells us that the head(s) have an inalienable possessor, but it does not directly encode this relation through binding by the subject of HAVE. We assume that instead, this interpretation arises through a pragmatic inference. With Vergnaud and Zubizarreta (1992), we take it that a distributive reading obtains when the NP is interpreted as a type. Once again, this is not something that is determined by the grammar, but it is rather the fruit of an inference. All other aspects of this structure are as above (cf. the tree in (28)). In particular, the argument position corresponding to that of the agent of *uelare* ‘veil’ is similarly existentially closed at SpecVoiceP: as a result, the example is compatible with a reading in which the gods veiled themselves, as well as with one where the agent of the veiling event is some third party.⁵³

⁵³The non-coreferential reading seems to be the most prominent one for this particular example. The gods referred to are the statuettes of the Penates which Aeneas brought with him from Troy. In Vergil’s *Aeneid* (3.147-178A), the latter come to life in Aeneas’ dream, and the issue that Servius mentions is the fact that they only wear a head-band, no veil which covers their hair entirely. Clearly, although in the scenario in which the statues have come to life, it is possible to portray them as volitional agents who put on their own

Three alternative analyses can be ruled out for the following reasons. First, should type-denoting nouns be systematically indexed (viz. with a variable index), they would always need a binding category, but as we saw above, Latin external possessors may be omitted, even in cases that cannot plausibly be analysed in terms of *pro*-drop (see in particular (41d)). Second, if token-denoting NPs carried an index (a constant), it would not be clear why this indexing mechanism would end up being interpreted radically different in Latin and in French (recall that in the latter language, NPs with a constant index always denote types, even if those types later come to instantiate a token, viz. at the DP-level, along lines described above). Third, one could hypothesize that rather than being existentially closed, the understood possessor argument of the relational noun *caput* ‘head’ is in fact realized as a null PRO in SpecNP. If so, the reference of this empty category would have to be established through binding by the NP *dii* ‘gods’. This does not seem possible, due to the presence of an intervening argument position, viz. the lower SpecVoiceP where existential closure of the understood Agent of *uelare* is encoded. We return to the possibility for relational nouns to have an implicit PRO external argument in section 5.2.

None of these problems arises if we assume that in the earlier stages of Latin, both the type-token distinction and that between alienable and inalienable possessors were not encoded structurally, but rather inferred pragmatically.

5 Romance DP syntax and the innovation of perfects

In this section, we present our analysis of how a change in the functional make-up of argument nominals triggered the grammaticalization of true perfects.

5.1 From demonstratives to definite articles

The Romance languages all have grammaticalized definite determiners, which are typically derived from the Latin distal demonstrative pronoun *ILLE* (more rarely, they derive from the reflexive intensifier *IPSE*: this is the case in, for example, Balearic Catalan).⁵⁴ But how exactly did the change from demonstratives to definite articles unfold?

5.1.1 Stages in the grammaticalization of definiteness

There are several models of the grammaticalization of definite articles (see, e.g., De Mulder and Carlier 2011; Skrzypek et al. 2021). Here we will use the proposal of Skrzypek (2012, 49), which has the advantage of breaking down the development from demonstrative determiner to definite article into several steps. A summary of the model we will assume is given in (45):

- (45) a. Stage I: (distal) demonstrative (deixis)
 b. Stage II: direct anaphoric marker (textual deixis)

headgear, it seems more appropriate to hold the sculptor responsible for not having represented as veiled the artefacts they produced.

⁵⁴For further discussion of the Latin prehistory of the Romance definite articles, see Väänänen (1981, 121), Selig (1992) and Adams (2013, 482–527).

- c. Stage III: indirect anaphoric marker (textual)
- d. Stage IV: uniqueness marker (non-textual)
- e. Stage V: generalized article (also for generics; non-textual)

In Stage I, a pure demonstrative has to refer to some entity in the speech situation, as in (46a). In contrast, a Stage II article has to refer back to the same textual antecedent, as in (46b). In Stage III, the textual antecedent of a definite DP does not have to take the shape of a coreferential category: rather, it can also be an entity from which the existence of the relevant definite DP can be deduced. This usage is referred to as *bridging* or *associative anaphora*. In the prototypical case, an associative anaphor and its antecedent stand in a part-whole relation: this is illustrated in (46c), where *the church* is most naturally interpreted to be that of the village newly introduced in the previous sentence, not some other church evoked in earlier discourse. Next, a Stage IV definite determiner no longer needs a textual antecedent, not even an indirect one: this is illustrated in (46d). While we are not completely sure about which criteria may identify a Stage V definite determiner, (45e) is a promising candidate, since it features a definite determiner expressing genericity.

Carlier and De Mulder (2010, 249) have argued that associative anaphora interpretations (that is, Stage III definites) are not available for demonstrative determiners. In other words, convincing cases of associative anaphora involving a definite determiner can be construed as trustworthy evidence that grammaticalization has reached a stage with *bona fide* definite articles.

- (46)
- a. *This dog* (here in front of me) is dirty.
 - b. He brought a cat and a dog. *The dog* was dirty.
 - c. He came to a village. *The church* was splendid.
 - d. *The sun* was shining.
 - e. *The whale* is a mammal. [generic reading]

Before we turn to the historical development of definite articles in Latin/Romance, two important remarks are in order. First, the absence of an overt article should not be taken to reliably indicate the absence of DP-syntax. In other words, articleless nominals may actually contain a D-layer, whose precise content may vary: (i) with Longobardi (1994) we may assume that lexical nouns may head-move to D, an operation which prevents the insertion of an article in the same position; (ii) alternatively, D may contain a phonologically null (definite) determiner (Contreras 1986). Second, the stages listed in (45) do not feature expletive articles of the type we discussed in section 3.1. We are not aware of any clear indication of where expletive determiners should appear in this pathway of change. It seems reasonable to assume that definite articles in French examples like (29) should be classified as at least as Stage III definites, as they strongly resist being replaced with a demonstrative determiner. However, we are not in a position to exclude that the availability of these uses might actually be indicative of an even more advanced stage of grammaticalization (IV, or even V).

5.1.2 Definite articles from Late Latin to Romance

There is a wide consensus in the literature that true definite articles can be found in texts dating from as early as the eighth century (for an overview of early text examples featuring definite articles from different regions of the *Romania*, as well as further bibliographical references, see Varvaro 2013, 31–32). A particularly relevant example is given in (47), where three unambiguous occurrences of the definite article are underlined (on this example, see, among others, Beckmann 1963, Avallé 1965; Norberg 1968, 28; Väänänen 1981, 198–199; Putzu and Ramat 2001, 112; Kiesler 2018, 144–147):

- (47) & ipsa cuppa frangant la tota, ad
 and DET.F.SG cup.(F.)SG break.PRS.SBJV.3PL PRO.F.SG entire.F.SG to
 illo botiliario frangant lo
 DET.M.SG wine_steward.(M.)SG break.PRS.SBJV.3PL DET.M.SG
 cabo, at illo scancio tollant lis
 head.(M?.)SG to DET.M.SG cup_bearer.(M.)SG take.PRS.SBJV.3PL DET.F.PL
 potionis [...] Cum senior bibit duas uicis,
 drinks.(F.)PL when lord.(M.)SG drink.PRS.3SG two.F.PL time.(F.)PL,
 sui uasalli la tertia bonum est
 his.M.PL vassal.(M.)PL DET.F.SG third.F.SG good.N.SG be.PRS.3SG
 ‘And that cup, let them break it completely, and let them smash the head
 of the wine steward, and let them take the drinks from the cup-bearer [...]
 When the lord drinks twice, the third <time> his vassals may drink too.’
 (Parody on the *Lex Salica* 12, ca. second half of the 8th c. CE, text from
 Väänänen (1981, 198), ll. 8–11)

Note that the first underlined occurrence involves inalienable possession, and that both the first and the second underlined tokens represent associative anaphors.⁵⁵ In the classification of Skrzypek (2012), these are thus Stage III definites, and according to Carlier and De Mulder (2010), they cannot be demonstratives anymore - which would be the alternative and ancestral value for *lo* and *lis*.⁵⁶ Finally, the third underlined occurrence involves a case that must be interpreted as a Stage IV definite, since *the third N* is necessarily semantically unique (as would be the case for superlatives), and its reference does not depend on the specific context.

Although in the earliest stages attestations are very sparse, the broad outlines of the subsequent spread of definite determiners in the Romance languages may be

⁵⁵ *Lis potionis* ‘the drinks’ could in principle be a Stage II direct textual anaphor, but there is no plausible antecedent in the preceding context; an associative anaphor related to the cupbearer seems the most likely interpretation.

⁵⁶ A reviewer questions our position as being very (too) conservative with respect to recognizing early true perfects, and overly permissive when it comes to recognizing early attestations of true definite articles as genuine. We do not see a contradiction here: we can explain away all claimed early instances of true perfects, but not (47) as an instance of a true definite article. That being said, do examples like (47) constitute incontrovertible evidence that already in the 8th century, there were type-denoting definite articles of exactly the same kind as (29b) in contemporary French? We cannot be entirely sure. For instance, we would need to see more evidence of the scopal behaviour of definite determiners with respect to a universal or distributive quantifier.

reconstructed as follows. On the one hand, we may assume that during the transition from Latin to early Romance, there was a period of grammar competition (in the sense of Kroch 1989, 1994), with the old NP-grammar and the innovative DP-grammar co-existing for at least some centuries.⁵⁷ On the other hand, the endpoint of the development is the present-day situation in which nominal arguments are either (i) DPs with an overt determiner, (ii) DPs with a covert determiner (Contreras 1986; Chierchia 1998), (iii) DPs without a determiner but with N-to-D movement (Longobardi 1994) or (iv) bare NPs which radically lack a D-layer. Importantly, the last three of these four categories cannot be distinguished from each other on simple morphosyntactic grounds (they all surface as NPs occurring without an overt determiner). In what follows, we will use the term ‘determinerless NP’ to refer to this surface pattern. NPs without a D-layer (that is, only case (iv)) will be referred to as ‘bare NPs’. What the underlying syntax is of any given determinerless NP has to be inferred from the syntactic and semantic properties of the relevant item (for example, N-to-D movement implies a rigid denominator (proper name) or reference to kinds, whereas null articles are referentially less constrained). This is entirely parallel to DPs with an overt definite article, whose interpretation is also not *a priori* fixed (the definite determiner may be contentful (strong) or expletive (weak)).

Setting aside intermediate stages in historical Romance for the time being, let us first concentrate on the present-day situation. Across the Romance language family, the distribution of the four types of nominal arguments mentioned above is subject to a fair amount of (micro-)variation (Dobrovie-Sorin and Laca 2003; Espinal and McNally 2011; Cyrino and Espinal 2015), but some generalizations can be made. In particular, there are good reasons to believe that bare NP arguments are not remnants of the old NP-grammar inherited from Latin. The most important argument in favour of this claim is related to the highly restricted syntactic and semantic properties of this type of nominal. Specifically, setting aside determinerless nominal complements to prepositions, bare NP arguments can only ever appear as direct object; from a semantic point of view, the relevant items do not refer to entities but properties.⁵⁸ In their discussion of determinerless arguments in (spoken varieties of) Brazilian Portuguese, which among the Romance languages is most liberal in allowing for argument nominals to appear without an overt determiner, Cyrino and Espinal (2015) suggest that truly bare NP objects do not qualify as genuine thematic arguments, but rather as predicate modifiers.⁵⁹ They (p. 477) furthermore observe that what appear to be bare NPs functioning as genuine entity-denoting arguments are necessarily interpreted as definite: from this observation, they infer that these determinerless NPs are in fact DPs (with a null determiner or with N-to-D movement). As a result, and despite surface counterexamples, it may after all be correct that across the Romance language family, nominal arguments must necessarily contain a D-layer (Longobardi 1994; Chierchia

⁵⁷See Simonenko and Carlier (2020) for a generalized grammar competition account of the spread of definite articles in Old French.

⁵⁸Espinal and McNally (2011) show that in Catalan, bare NP objects always take narrow scope, that they are number-neutral, and that they are extremely limited in their capacity to allow for discourse anaphora with personal pronouns: they propose that these objects are property-denoting and that they are (pseudo-)incorporated.

⁵⁹For further discussion of the syntax and semantics of determinerless NPs in Brazilian Portuguese, see Schmitt and Munn (2002), Müller and Oliveira (2004) and Lima and Thomas (2022).

1998). In any event, for reasons that will become clear in the following section, in the present context the main issue is whether or not a given nominal argument has a syntactically projected D-layer; the lexicalization of the latter (null or overt) is of lesser importance.

Let us then return to the situation in early Romance: we will take Old French as an example, as the spread of definite determiners in this language is fairly well understood (cf., e.g., Carlier and Goyens 1998; Mathieu 2009; Déchaine et al. 2018). At a descriptive level, we see that nominal arguments with and without an overt determiner coexist, and that the interpretation of the variant without article is not restricted to a particular semantic class of referents: the latter can have definite, indefinite and kind reference, and they are compatible with both count and mass nouns. Importantly, as shown in Déchaine et al. (2018), the incidence of determinerless arguments is not random: rather, in a quantitative corpus study with a multi-factorial design, nominal arguments without a determiner can be shown to be favoured (i) when acting as the direct object, (ii) when appearing postverbally, (iii) with non-count nouns (abstract entities and mass terms), (iv) with indefinite reference, (v) with plural nouns and (vi) with feminine nouns.

With Déchaine et al. (2018), we will assume that by the Old French period, a phonologically null determiner had come into being. Initially, this option may have alternated with bare NP arguments, yielding surface ambiguities which are very difficult to resolve on the basis of corpus data. However, we take it for granted that null determiners (perhaps alongside N-to-D movement) were part of the grammar of Old French.

In any event, there is every reason to believe that the genesis of grammaticalized definiteness predates the innovation of HAVE-perfects (on the chronology of the latter, see section 2.1). In what follows, we will capitalize on this observation by positing a causal connection between the rise of definite articles and that of periphrastic perfects with HAVE.

5.2 Innovating perfects

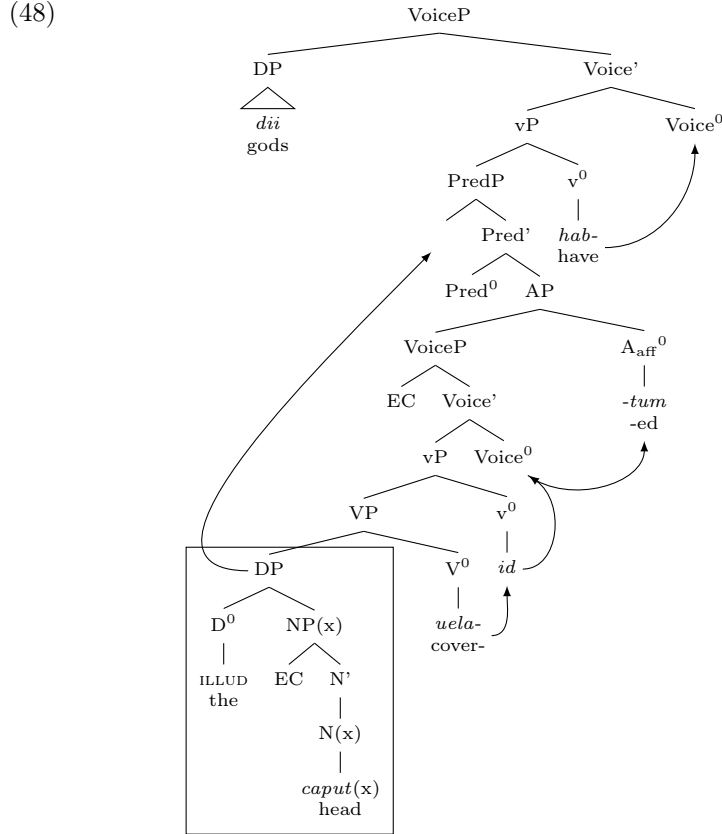
We are now finally in a position to spell out how a very specific subset of Late Latin HAVE-statives is predicted to be reanalysed as true perfects. Our crucial contention is that structures like (44) are not compatible with the newly innovated DP-grammar (with articles that have at least reached stage III of the grammaticalization path shown in (45)). Given the presence of a D-layer above N, the type/token distinction in the nominal domain came to be encoded in the syntax (along lines of Vergnaud and Zubizarreta (1992)). As a consequence, L1-learners with an innovative DP grammar confronted with a certain type of input from conservative NP-grammar speakers had no choice but to innovate true perfects.

We will proceed as follows: in section 5.2.1, we will show why the old configuration could no longer yield an acceptable parse once a D-layer had been introduced into the grammar. Next, in Sections 5.2.2 and 5.2.3, we discuss two ways in which the new grammar could attribute acceptable parses to the ‘problematic’ input, namely an elimination of all verbal functional structure on the one hand, and the development of true perfects on the other. Since the conditions under which in particular this

last reanalysis is assumed to have taken place are highly specific, in section 5.2.4 we will consider whether the relevant contexts were sufficiently frequent in (very) Late Latin in order for our proposed scenario to be plausible. In section 5.2.5 we briefly consider how the newly created transitive HAVE-perfects may have spread to other environments.

5.2.1 The emergence of a D-layer and its consequences

Consider a variant of the tree in (44), where the internal structure of the direct object is enriched with a D-layer. For the sake of concreteness, we lexicalized the head position of this projection as *ILLUD*, but the exact phonological shape of the newly coined definite determiner is immaterial (it may be overt or covertly realized):



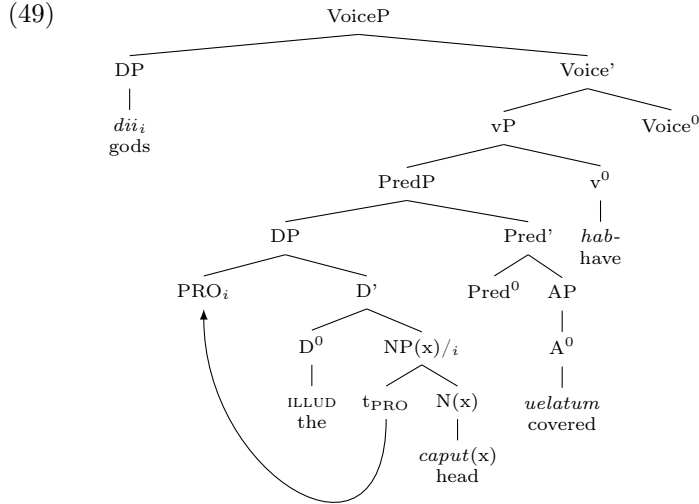
To obtain the desired inalienable interpretation and the associated distributivity effect due to the presence of a plural external possessor, we have to assume that in the new grammar, only *N(P)* is indexed, and that the determiner is of the expletive kind. Otherwise, if *D* were to induce token-reference, we would end up with a single token of a head, that could not be distributed among the plurality of gods. The problem is that there is no appropriate DP antecedent – phonologically explicit or otherwise, see below – to bind the variable of the relational noun *caput* ‘head’ (recall that at a semantic

level, this DP has to co-vary with the possessor argument of the inalienable possessee). One derivation that is not possible is direct binding by the externally merged state-holder of HAVE (namely *dii* ‘gods’), as the lower SpecVoiceP containing an empty operator inducing existential closure intervenes between the higher subject and the NP embedded in the direct object and could provide a suitably local binder. However, should such a local binding configuration be established, the resulting reading would not be the desired one: the structure here should be interpreted as a type of passive, where there is no commitment as to who veiled the heads (as mentioned, it need not be the gods themselves). With local binding at SpecVoiceP, the lower part of the structure would identify the agents of the veiling event(s) as inalienable possessors of the heads, and higher up the same heads would be said to be the alienable possession of the gods. Finally, the existentially closed possessor argument of the relational noun does not c-command the (highest occurrence of) the variable x in the tree, which sits at the level of the full NP: as a result, it does not qualify as a potential binder, regardless of its interpretive properties.

We see two possible ways in which a binding category could be provided, which both involve reanalysis of the original structure and indeed the innovation of a syntactic construction not available to earlier generations. We will examine them in Sections 5.2.2 and 5.2.3, respectively.

5.2.2 Repair #1: flattening the structure

The first new construction came about as follows. A structure like (48), once enriched with a definite article, is reanalysed in such a way that the participle becomes a plain adjective, devoid of a Voice-layer. Moreover, the understood possessor of the relational noun is now projected in the syntax, taking the shape of a PRO-subject first merged in SpecNP (a thematic position), which undergoes A-movement to SpecDP. In this way, a binder for the relational noun’s variable is provided, without there being any potential intervening binding category. Concretely, we take it that the adjective sits in the complement position of a PredP, and that in this predicative nucleus the PRO-subject and the relational noun are in a sufficiently local configuration in order for variable binding to be established. PRO itself is referentially bound (‘controlled’) by the external argument of HAVE (in SpecVoiceP).



This structure corresponds to Old French (6b) and Modern French (7b), which are both fairly close in meaning to Latin HAVE-statives. In fact, this repair scenario presumably does not involve any syntactic innovation beyond the possessed DP: rather, both the input and the output (the latter being structurally simpler) can be argued to have roughly the same syntactic structure, viz. that of a HAVE-stative.

Before we move on, let us add a couple of remarks about the nature of PRO as the subject of a DP (on this possibility in the particular case of constructions with an inalienably possessed object, see also Guéron 1995, 198). First, from a diachronic point of view it is important to note that in this particular structure, the option of a PRO-subject was not available earlier, given the absence of an appropriate specifier position in which PRO could act as the binding category of the variable associated with NP. Recall from section 4.2 that we take D to be the nominal correlate of clausal T: by this token, SpecDP can be said to qualify as an A-position for subjects in the nominal domain (for evidence that SpecDP counts as an A-position in present-day Italian, see Cinque 2014). Note also that our analysis allows us to assume that the lexical entry of a relational noun like *caput* ‘head’ remains constant over time, in that at all stages, it is subcategorized for an obligatory possessor argument. As indicated in the tree in (49), PRO is base-generated in SpecNP, taking the place of the earlier empty operator which caused the possessor argument to be interpreted as existentially closed. The new structure is minimally different, in that the identity of the possessor is no longer purely pragmatically established, but rather encoded in the syntax, by means of a control relation.⁶⁰ Furthermore, we take it that A-movement

⁶⁰Arguments in favour of syntactically representing implicit arguments of nouns comes from binding facts such as those illustrated in (i) and (ii) (from Bhatt and Pancheva (2017, 16), their (52)):

- (i) *They_i told [PRO_i stories about them_i].
(ii) *[The PRO_i realization [that John_i was sick]] upset him_i.

In (i), unacceptability (given the coreference relations indicated) can be argued to be due to the presence of an implicit agent argument (represented here as ‘PRO’) associated with the noun *stories*: specifically, the combined facts of this element c-commanding the pronoun *them* and these two items sitting in the same

of PRO to SpecDP results in a configuration in which the possessor asymmetrically c-commands the index of NP, with successful binding of the latter as a result.⁶¹ Note finally that the insertion of PRO in SpecNP does not eliminate the need for NP itself to be endowed with a variable index: not only would omitting this variable amount to creating another source of parametric variation regarding the encoding of inalienable possession, it would also become unclear how the relational noun would end up having type-reference.

Second, an alternative analysis involves base-generating the possessor inside the DP headed by the relational noun, and subsequent movement of this element to SpecVoiceP, yielding a type of possessor raising. We believe this option should be discarded for the following reasons. First, this operation would violate locality, in that a putative Voice-Probe would attract a DP embedded in another DP, whereas nothing in the feature composition of both DPs involved seems to justify this type of subextraction (for example, it is not the case that the more deeply embedded DP has an additional feature which allows it to by-pass the DP dominating it).⁶² Second, possessor raising to SpecVoiceP would involve movement into a theta-position, which is widely considered to be illicit (as it violates the Theta Criterion, which postulates a one-to-one relation between thematic roles and arguments).

In any event, the strategy of structural reduction yielding configurations like (49) is not always available, because not every passive PaPa is eligible for reanalysis as a plain adjective (without any verbal functional structure). More specifically, a second type of reanalysis must obtain when a HAVE-stative contains (i) a relational noun with a D-layer in need of a binding category, and, crucially, (ii) a past participle which cannot be analysed as a plain adjective. As we will proceed to show, the second of these conditions obtains unambiguously in cases where a single verbal root is associated with more than one deverbal adjective, to the effect that one form has become specialized as a pure (lexicalized) adjective, the other being a verbal PaPa.

5.2.3 Repair #2: innovating true perfects

Consider the example in (50) (in which the HAVE-verb appears as a present participle):

local domain amounts to a violation of Principle B of the Binding Theory. Similarly, in (ii) the fact that the implicit experiencer of the noun *realization* c-commands the coreferring R-expression *John* contravenes Principle C.

The fact that the example in (iii) is interpretively ambiguous has been argued to indicate that implicit arguments of nouns are only optionally projected in the syntax (at least with certain nouns). This example, with binding of the anaphor *each other* by the matrix subject *they*, is grammatical, irrespective of who was telling the stories. This is unexpected if an agent argument disjoint in reference with the matrix subject were to be obligatorily projected in SpecDP, given that this element would block binding of the anaphor by the subject of the finite clause, contrary to fact.

(iii) They_i heard [PRO_{i/j} stories about each other_i].

However, we may assume that in a case like (iii), the agent may either be existentially closed or realized as a null pronoun. Only the latter would be able to participate in processes of pronominal binding. If this analysis is on the right track, the diachronically variable realization of internal possessors of relational nouns in Latin/Romance argued for in the main text would be largely parallel to the English facts just reviewed.

⁶¹It seems reasonable to assume that the pre-existing existential operator is not endowed with the appropriate syntactic features required to similarly undergo A-movement.

⁶²On subextractions of this last type, see Rizzi (2016).

- (50) qui [lotas aqua manus
REL.NOM.M.PL washed.ACC.F.PL water.ABL(.F).SG hands.ACC(.F).PL
habentes] conscientiam liuore
have.PRS.PTCP.NOM.M.PL conscience.ACC(.F).SG envy.ABL(.M).SG
pollutam gestabant
sullied.ACC.F.SG carry.IPFV.3PL
‘who, having washed their hands with water, still had their conscience sullied
with envy’ (Beda Venerabilis, *In Marci euangelium expositio* 2.7, early 8th c.
CE)

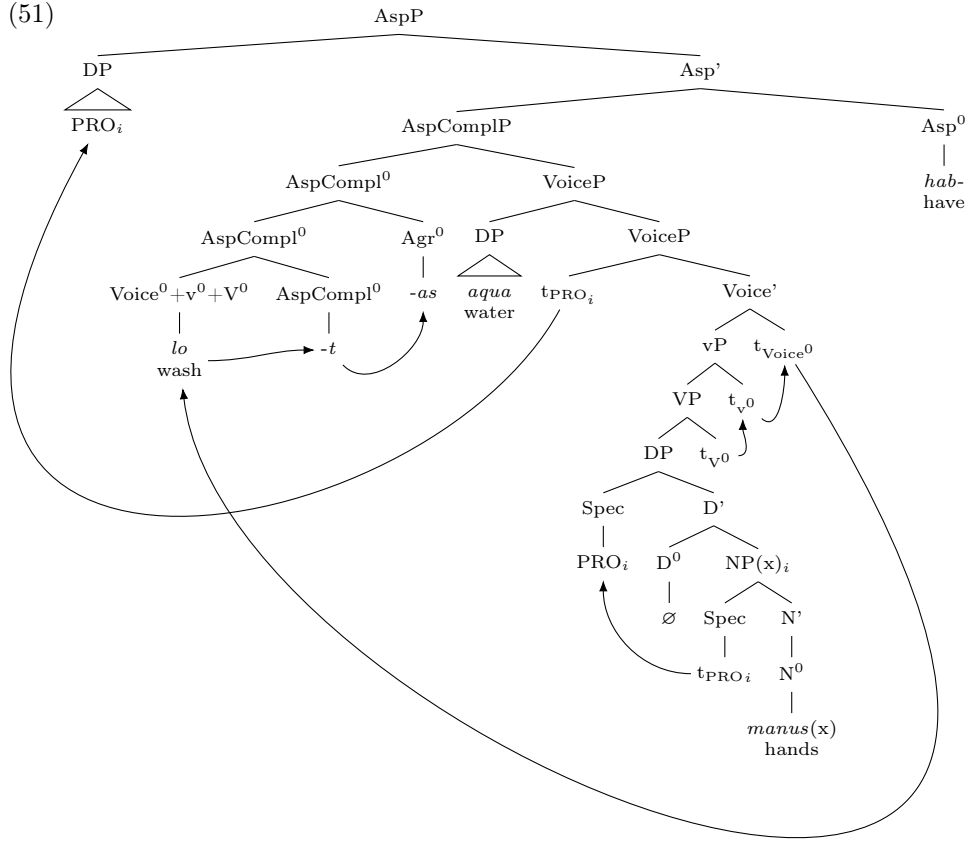
Let us go through the main features of this example. First, the predicate of the HAVE-stative in (50) is a form of *lotus*, that is, the past participle of *lauo* ‘wash’. As is well known, this participle exists alongside a competing deverbal adjective derived from the same root, namely *lautus*. Crucially, as noted in Hey (1890, 143) (see also Beikircher 1973, 1053), variation between the two forms is not merely phonological in nature, but rather, by the Late Latin period the two variants had specialized along a semantic dimension, with *lotus* acting as the true past participle of *lauo*, and *lautus* as an adjective meaning ‘clean’, and by extension ‘fine, sumptuous, brilliant’. Second, the participle *lotas* is modified by the ablative phrase *aqua* ‘with water’: though this may well be a pseudo-incorporated modifier (in the sense of Gehrke 2015) rather than a true instrumental adjunct, its very presence is indicative of at least some verbal functional structure which is absent from purely adjectival projections. Third, the subject of the small clause is the relational noun *manus* ‘hands’, and the intended reading is one with an inalienable possession relation (the external possessor being the PRO-subject of the participial clause, whose reference is in turn controlled by the relative pronoun *qui* ‘who’). Finally, we take it that although this example is in principle ambiguous (as a result of the agent argument of *lotas* being existentially closed, along lines explained above), the reading in which the subjects referred to have washed their own hands is pragmatically more plausible than one in which the agent of the washing event is some third party.

The innovation of true periphrastic perfects may then be modeled as follows. A possible tree for the example in (50) is given in (51).⁶³ Let us assume a slightly idealized scenario in which (50) was produced by a language user whose mental grammar does not have definite articles, and that this string is to be parsed by a member of the generation of language learners who have newly innovated this category. Recall that for this first generation of article users, there must have been competition between the old NP-grammar and the innovative DP-grammar, and that in the latter system, all nominal arguments are endowed with a D-layer, with or without a phonologically overt article. As a result, two parsing options are available for the HAVE-stative in (50): either the NP-grammar is selected, and no reanalysis takes place. Alternatively, the DP-grammar is selected: as a result, the noun *manus* ‘hands’ comes with a phonologically null expletive article, and this entails the need for the inalienable possession

⁶³For reasons of space, in (51) the structural representation of the agreement morphology on the past participle in the complex AspCompl-head is simplified; we have also dispensed with the entire functional structure above AspP. We take AspComplP (i.e., completive aspect) from Cinque (1999, 100) to be a functional projection above VoiceP and below AspP proper.

relation to be encoded in the syntax.⁶⁴ As above, a PRO-subject carrying the thematic role of possessor is generated in SpecNP: after it has A-moved to SpecDP it can bind the variable associated with NP. However, what is crucially different is the way in which PRO is controlled: assuming that any subject position projected by *habere* sits too far away to act as a controller due to the intervening Voice-layer, the only way in which a sufficiently local antecedent can be provided is by projecting an agent DP in the Voice projection on top of the participle *lotas* (in this case, given that the HAVE-stative is headed by a present participle, this also happens to be PRO). Importantly, in this structure, the subject of the participle and the subject of the HAVE-verb are obligatorily identified.

⁶⁴There is interesting evidence from L1-acquisition suggesting that expletive articles do not pose any learning difficulties, quite on the contrary (Baauw 2002). On the one hand, learners of Spanish, a language that uses expletive articles in inalienable possession constructions, master the use of definite determiners (including those occurring in distributive inalienable contexts) in an adult-like fashion around the age of 5–6. On the other hand, learners of Dutch, which lacks expletive articles, go through a non-adult-like stage in which they have access to the expletive interpretation of definite determiners.



We consider this structure to be a true, monoclausal HAVE-perfect by virtue of the fact that it contains only a single VoiceP. This effectively derives the obligatory coreference between the external argument of the participle and the subject of *habere*: the two are simply one and the same syntactic entity. If, on the other hand, both the PaPa and the HAVE-verb were to have their own Voice-layer, we see no way in which the obligatory coreference between the (full or existentially bound) arguments they introduce could be forced. Specifically, any alternative structure with two VoicePs is problematic with respect to the external argument of the lower VoiceP. If the latter is existentially closed, the obligatory coreference cannot be deduced. If both hypothesized VoicePs were to share a single argument, we would have to assume this to be introduced in the lower VoiceP, later to undergo movement to the higher one, which would instantiate movement into a theta-position: as mentioned, we consider this type of movement to be impossible. A final option is for a hypothesized lower VoiceP to have a PRO-subject, referentially controlled by the external argument of *habere*; however, the only participial clauses we are aware of which have a PRO-subject are adjuncts (non-finite adverbial clauses or adnominal reduced relatives), which renders this option unlikely.

That being said, it is clear that the structure in (51) is not yet identical to the various HAVE-perfects that we know from the Old and Modern Romance languages. On

the syntactic side, many subsequent word order changes gave rise to the emergence of exclusive head-initial orders ('HAVE-PaPa(-object)'), and in some Romance varieties (like Spanish) also an adjacency requirement between the auxiliary and the participle. Semantically, a series of secondary grammaticalizations led to the emergence of various readings of the periphrastic perfect.

5.2.4 How frequent was the critical environment?

The particular syntactic context which we assume meets all three necessary conditions that make it a suitable input string for reanalysis is clearly a very specific one; given our assumption that L1-learners were responsible for reanalyzing the relevant strings as periphrastic perfects, it is useful to consider whether the token frequency of the surface string which we deem crucial for this development was high enough for this scenario to be plausible. Two considerations lead us to believe that this is indeed the case. First, we carried out a small corpus study which shows that HAVE-statives with an inalienable NP.ACC and an agentive PaPa (with the required coreference relations) do indeed occur at non-negligible rates. We took into consideration a selection of texts ranging from the early fourth until the late sixth century CE.⁶⁵ From this corpus, we extracted all forms of the lexeme *habere*, yielding a total of 4,773 occurrences. The most important quantitative tendencies we observed are summarized in Table 1.

In 180 cases out of a total of 4,773 occurrences, *habere* appears with a small clause complement in which the predicate is a PaPa (which corresponds to 3.77%); despite there being some variation between individual authors, the relevant tokens are fairly evenly distributed across the entire corpus. Of those 180 HAVE-statives, 108 are compatible with a reading in which the state holder of *habere* is coreferential with the understood agent of the PaPa. This is the set of instances of *habere* that Latin linguists generally agree to qualify as potential sources of true perfects. In contrast, the subset that we identify as the right environment in which HAVE-perfects originated is that of 12 tokens in which the subject of the small clause is an inalienably possessed noun.⁶⁶ In other words, our proposal is more restrictive than the traditional theory: specifically, in our account only around 11% of the cases that are traditionally taken to be relevant qualify as potential input strings for reanalysis. We are not aware of

⁶⁵The following texts were taken up into the corpus (unless mentioned otherwise, the complete preserved works of the relevant authors were taken into account; (approximate) dates are provided in table 1): Lactantius (*Divinae Institutiones* (supplemented by the *Epitome diuinarum institutionum*, *Constantino dedicationes* and *Dualistici loci*), *De ira Dei*, *De opificio Dei*), Palladius, Ambrose (*De paradiso*, *De bono mortis*, *Epistulae*), *Itinerarium Egeriae*, Jerome (*Epistulae* 1–106), Ammianus Marcellinus, Augustine (*Confessiones*, *Contra Iulianum*, *De beata uita*, *De dialectica*), Acts of the Council of Carthage (411 CE), Vegetius, Cassius Felix, Apollinaris Sidonius, Victor of Vita, Caesarius of Arles (*Sermones* 1–116), Iordanes, Gregory of Tours (*Historiae*, *Liber in gloria confessorum*, *Liber in gloria martyrum*, *Liber de uirtutibus Martini*, *Liber uitae patrum*) and Gregory the Great (*Moralia in Iob praefatio*–18.41).

⁶⁶The relevant 12 examples are the following: Ambrose, *De paradiso* 13.63 (*clausos oculos corporis habuit Adam*), 14.70 (*deus clausos oculos habebat*), *Epistulae extra collectionem traditae* (CPL 0160 1 (M)) 1.12 (*ne pollutam habeas conscientiam*); Jerome, *Epistulae* 18A.12 (*habebant peccatorum sordibus pollutos pedes*), *Epistulae* 77.5 (*dissuta habuit latera*); Vegetius, *Digesta artis mulomedicinae* 1.4 (*nares habet extensas*), 2.39.3 (*maxillam sideratam habuerit et tortam*), 2.92.1 (*collum habebit extensum*), 2.94 (*nares tensas habebit*), 2.113.1 (*cervicem inclinatam habebit*); Victor of Vita, *Historia persecutionis Africanae provinciae* 2.11 (*linita habentes stercoribus uestimenta, facies simul et capita*); Gregory of Tours, *Historiae* 6.40 (*oppilatas habeas aures*). To be sure, these 12 examples are most naturally interpreted as statives, because that is what they are (reanalysis as HAVE-perfects has not yet happened). The important points are (i) that the relevant tokens are in principle structurally ambiguous, and (ii) that their truth conditions are sufficiently compatible with those of string-identical true perfects for reanalysis to be contextually plausible.

Author	Date	<i>habere</i>	HAVE-stative	Coref.	Inal.	Coref. + inal.
Lactantius	300	541	19	14	1	0
Palladius	350	62	6	3	0	0
Ambrose	380	577	30	16	4	3
<i>Itinerarium Egeriae</i>	385	53	3	2	0	0
Jerome	390	555	24	13	5	2
Ammianus Marcellinus	395	52	2	1	0	0
Augustine	410	570	10	6	1	0
<i>Acts o/t Council of Carthage</i>	411	313	4	3	0	0
Vegetius	420	164	16	10	10	5
Cassius Felix	447	27	0	0	0	0
Apollinaris Sidonius	470	70	4	2	0	0
Victor of Vita	490	51	3	3	1	1
Caesarius of Arles	520	600	12	4	4	0
Iordanes	550	66	4	4	0	0
Gregory of Tours	590	521	38	25	6	1
Gregory the Great	595	551	5	2	2	0
TOTAL		4773	180	108	34	12

Table 1 The frequency of Late Latin HAVE-statives with an inalienable NP.ACC

any theory that predicts how many input tokens an L1-learner needs in order to bring about syntactic change. Nevertheless, our corpus data suggest that there is no reason to believe that the environment we deem crucial for the genesis of HAVE-auxiliaries was exceedingly rare.

Our second argument in support of the claim that the scenario we have sketched is plausible is more conceptual in nature. The corpus data that we just analysed are arguably not entirely representative of the type of input that an L1-learner of (Late) Latin may have been exposed to; arguably, the data we do have lead to an underestimation of the actual frequency of the input string that we are interested in. Unfortunately, we do not have access to child-directed speech of the relevant period (and it is indeed doubtful that we ever will). Nevertheless, in the corpus data that we have at our disposal, it is striking that many examples meeting all three necessary conditions identified at the beginning of this paper tend to share a number of characteristics. Specifically, NP.ACC is typically an inalienably possessed body part, most often a form of the lexemes MANUS ‘hand’, CAPUT ‘head’, PES ‘foot’ or OCULUS ‘eye’; less frequently, it is a noun denoting an abstract property (like a mental faculty) or a concrete object belonging to an individual’s personal sphere, such as a piece of clothing. Conversely, the PaPa tends to be derived from a grooming verb (particularly *lauare* (or *lauere*) ‘wash’). Interestingly, there is ample evidence that these lexical items have a high token frequency. First, nouns referring to body parts are standardly considered to be part of a language’s core vocabulary. For example, 19 of the 100 elements in the ‘Leipzig-Jakarta list of basic vocabulary’ denote body parts (Tadmor 2009, 69–71). Second, in the particular case of Latin, the lexemes CAPUT ‘head’ (200th) and OCULUS ‘eye’ (203rd) feature among the 250 most frequently used words in a corpus of literary texts from the Archaic and Classical periods (Denooz 2010). It seems likely that the token frequency of nouns denoting body parts was even higher in child-directed

speech than in the texts considered by Denooz (2010) (historiographical prose, epic poetry, *etc.*).

To conclude, the evidence we have reviewed suggests that the syntactic context that we single out as the crucial one for the reanalysis of Latin *habere* as a TAM-auxiliary was well represented in actual usage.

5.2.5 From transitive to intransitive perfects

Once (resultative) HAVE-perfects were innovated in the particular context described above (viz. that of a transitive verb with a relational DP [itself with a null expletive article] as its internal argument), there is no principled reason (conceptual or empirical) why the same construction should not be possible in other contexts too: in particular, we predict that HAVE-perfects should also be available for intransitive verbs, and in the case of transitive verbs, also with non-relational DPs acting as the direct object.⁶⁷ As documented in Herzog (1910, 177–185), true intransitive HAVE-perfects are indeed attested in the early stages of the individual Romance vernaculars (see, for example, the Old French example in (52a), and the one from Old Italian in (52b)):

- (52) a. fors soul le lit o il at geut tant
 maybe on the bed where he have.PRS.3SG laid so_often
 ‘maybe on the bed he has laid so often’ (*Vie de Saint Alexis* 55.4 (ed. Paris), *ca.* mid 11th c.)
- b. tu non ai sognato
 you not have.PRS.2SG dreamed
 ‘you have not dreamed’ (*Cento novelle antiche* 37, late 13th c.)

Many early HAVE-perfects featuring a transitive verb were probably still ambiguous, also allowing for an interpretation as a HAVE-stative. However, with intransitives, this reading is not available.

To conclude, we may now answer the question as to why periphrastic HAVE-perfects were not innovated earlier: this question is pertinent in light of the fact that from Archaic Latin onwards, many surface strings were structurally ambiguous, and in principle compatible with a parse in which *habere* is an auxiliary. Two reasons for this come to mind. On the one hand, no trigger – like the need for a type-variable to be bound – existed which could force reanalysis. On the other hand, imagine that reanalysis took place without a trigger (by accident, so to speak). In that case, the output of reanalysis would always be near-indistinguishable from the pre-existing HAVE-statives like (44): this makes it extremely unlikely for such an innovation to spread across an entire speech community.

⁶⁷Note, on the other hand, that the innovation scenario proposed here is also compatible with one involving a ‘failed change’, that is, a change that never establishes itself as an integral part of the grammar, and which does not spread throughout a wider speech community. The reason why HAVE-perfects ended up being particularly successful across the Romance language family may be due to a variety of reasons, including contact with languages that had independently innovated the same pattern. On the role of language contact and other areal factors influencing the spread of HAVE-perfects across Indo-European, see Drinka (2017).

6 Conclusion

In this article, we pursued two aims: first, we provided a detailed analysis of the Latin source construction of Romance HAVE-perfects (with special reference to their syntactic structure). Second, we also proposed a new hypothesis of what caused the grammaticalization of Romance HAVE-auxiliaries. Our main claim is that the direct trigger of reanalysis was the emergence of the definite article in Late Latin.

In agreement with de Acosta (2011) and Târa (2014) (among others), we argued that the Latin construction ‘*habere* + NP.ACC + PaPa.ACC’ was not yet a full-fledged perfect, but rather a grammatically passive structure whose main verb is *habere*. Our analysis of the syntactic structure of these Latin HAVE-statives incorporates insights from Maienborn et al. (2016) (on the structure of different types of PaPa) and from Myler (2016)’s analysis of HAVE. We also presented evidence that the PaPa and its complement may denote event (and state) tokens, contrary to what has been described for stative passives in languages like German.

Building upon the analysis of inalienable possession by Vergnaud and Zubizarreta (1992), we further proposed that the emergence of definite articles (and thus the development of a functional D-layer) forced language acquiring children to reanalyse a particular subset of the pre-existing HAVE-statives as true perfects. Our point is that the particular context of inalienable possession *required* reanalysis of the source construction to take place. By this token, we claim to have identified the long-elusive trigger for reanalysis of a syntactic structure whose properties had remained stable for many centuries.

On the other hand, we remain, for the time being, uncommitted as to a potential cross-linguistic generalization positing a causal relation between the presence of definite articles and HAVE-perfects in a given language. On the one hand, the existence of definite articles seems to be no necessary condition for HAVE-perfects to arise. Witness the well-known fact that Hittite developed a perfect HAVE-auxiliary in the absence of articles. In other words, a surface-level generalization of the type “(only) languages with a definite determiner and HAVE-statives may develop perfect tenses” appears to be empirically inadequate. On the other hand, it is clear that the grammaticalization of definite articles (that is, the condition we have defined above as sufficient) should not automatically lead to the rise of HAVE-perfects. The history of Greek is interesting to consider in this respect, as the historical record displays an important time lag between the emergence of definite articles and that of HAVE-perfects. The latter are attested from the Homeric poems (8th c. BCE) onwards, where DPs with an overt article still alternate in many contexts with determinerless NPs. Conversely, unambiguous HAVE-perfects occur much later, perhaps as late as *ca.* 1600 CE (cf. Drinka 2017, 98; the early combinations of HAVE and a participle are restricted to transitive verbs, and are therefore likely to be more akin to what we have called HAVE-statives). Clearly, in order for the scenario we proposed for Latin/Romance to be replicated for Greek, all the necessary conditions we discussed need to be met too (there should exist HAVE-statives, and they should be able to occur with a small clause containing a ‘sufficiently verbal’ predicate and an inalienably possessed internal argument). Given the

rich body of textual evidence that is available for historical Greek, it should be possible to empirically verify whether the scenario we have proposed is viable for Greek too.

In sum, despite these caveats, we consider it worthwhile to explore the hypothesis that reanalysis of inalienable possession constructions embedded in HAVE-statives – crucially with an expletive definite article – may constitute a ‘privileged route’ towards HAVE-perfects.

More specifically, in future research it may be interesting to investigate whether there is, in any relevant sense, a ‘deep’, structural connection between definiteness marking and perfective HAVE-auxiliaries, to the effect that the historical case study analysed in this article would not be an isolated accident. For one thing, both definite articles and HAVE-perfects have been argued to be characteristic of ‘Standard Average European’ (SAE), that is, a set of isoglosses shared by a large number of languages spoken in Europe. In the literature on SAE, it is emphasized that many similarities among European languages are not merely due to historical relatedness, but also to many centuries of intensive language contact. A useful survey of the relevant data pertaining to definite articles and HAVE-auxiliaries can be found in Heine and Kuteva (2006) (see in particular their chapters 3 and 4). Although the authors do not posit an implicational (and indeed diachronic) connection between the two phenomena, some tantalizing patterns documented in this study merit further attention.

The Slavic language family is an interesting case in point. As noted in Heine and Kuteva (2006, 106), definite articles exist in languages at the northern and southern edges of the geographical area in which the Slavic languages are spoken, namely in North Russian on the one hand, and in Bulgarian and Macedonian on the other (see also Wiemer 2020, 256). In addition, there is evidence that the grammaticalization of demonstratives into definite articles is a change in progress in Polish, Sorbian, Czech, Slovenian, (Molise) Croatian and Kashubian, possibly as a result of contact with Germanic and Romance languages (Heine and Kuteva 2006, 112). The facts surrounding HAVE-perfects in Slavic are slightly more difficult to establish, but according to Heine and Kuteva (2006, 158) and Arkadiev and Wiemer (2020, 175–184), the languages in which the grammaticalization of perfective HAVE has progressed furthest are North Russian (on which, see also Kuteva and Heine 2004; Seržant 2012), Southern Thracian Bulgarian and in particular Southwestern Macedonian (on the history of HAVE-perfects in Bulgarian and Macedonian, see Hristov (2020)). The distribution of HAVE-perfects thus seems to be slightly more restricted than that of definite articles. This in turn suggests that Slavic languages could provide a further testing ground to evaluate the hypothesis that the scenario developed in the present paper for Latin/Romance may in fact be cross-linguistically not uncommon.⁶⁸

However, before any generalization of this type can be made, clearly additional information is needed about the respective timing of the grammaticalization of definiteness and HAVE-perfects in the languages concerned. Furthermore, care should be taken to disentangle language-internal developments from contact-induced changes. This last point is relevant for North Russian, whose history is marked by long-standing

⁶⁸On the possible relation between the presence of definite articles and HAVE-perfects in North Russian, see also Panzer (1984).

contact with German, and even more so for (varieties of) Bulgarian and Macedonian, which belong to the well-known ‘Balkan Sprachbund’. In the Balkan region, intensive language contact has famously led to a high degree of language convergence, including among phylogenetically unrelated (or only distantly related) languages, and importantly, both definiteness marking and HAVE-auxiliaries are among the features which have been argued to have become prominent in the relevant area as a result of pattern borrowing.⁶⁹

Investigating these issues in detail clearly goes beyond the scope of the present study, but we hope to have convinced the reader that the issues touched upon in the last couple of paragraphs merit further attention, and that they could shed more light on the historical developments that may lead to the genesis of HAVE-auxiliaries.

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⁶⁹For further discussion of the distribution of definite articles and HAVE-perfects in the Balkan languages, we refer to Tomić (2006) (in particular chapters 3 and 5).

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